MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
4000 PROCESS PLANT -LABORATORY BUILDING
DESIGN OF BUILDING STRUCTURE

Document Number **MD-216-49BL-EG-CV-CAL-1066**
Revision: **D01**
Revision Date: **19.10.2023**
Status: **AS BUILT**

Outotec Document Number:	900707	GAU01	AL01	CED01	69001	OU100688040	1
L&T Document Number:	O19045	300	C	K	24028	001	1

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	31.05.2020	FIRST ISSUE	MF	AVG	SJM	
C01	06.07.2020	ISSUED FOR CONSTRUCTION	MF	AVG	SJM	
D01	19.10.2023	AS BUILT	MF	AVG	SJM	

TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS 4
1. INTENT 5
2. EXECUTIVE SUMMARY 5
3. SCOPE 5
4. DESIGN METHODOLOGY & ASSUMPTIONS 6
4.1. Codes 6
4.2. Geotechnical investigations 7
4.3. Material Properties 7
5. DESIGN LOADS 7
5.1. Dead Load 7
5.1.1. Roof Dead Loads 7
5.1.2. Ground Floor Walls 9
5.2. Live Load 9
5.2.1. Roof Live Loads 9
5.3. Wind Load 9
5.4. Wind Load (Analytical Verification) 10
5.5. Seismic Load 13
5.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads) 13
6. DESIGN COMBINATIONS 14
7. CRACKED SECTIONS 15
8. DEFLECTIONS/DRIFT CHECK 16
8.1. Maximum Allowable Deflection 16
9. CONCRETE ROOF BEAM DESIGN 17
9.1. Design of Ledger Beam(LB1) 17
9.1.1. Staad Output For LB1 19
9.2. Design of Ledger Beam(LB2) 20
9.2.1. Staad Output For LB2 21

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.3. Design of Ledger Beam(LB3) 22
9.3.1. Staad Output For LB3 23
9.4. Design of Beam-(B1) 24
9.4.1. Staad Output For B1 25
9.5. Design of Beam-(B2) 26
9.5.1. Staad Output For B2 27
9.6. Design of Beam-(B3) 28
9.6.1. Staad Output For B3 29
10. CONCRETE GRADE SLAB DESIGN 30
11. DESIGN OF GRADE BEAM-(GB1) 31
11.1. Design of Beam-(GB1) 31
11.1.1. Staad Output for GB1 32
11.2. Design of Beam-(GB2) 33
11.2.1. Staad Output for GB2 34
12. CONCRETE COLUMN DESIGN 35
12.1. Design of Column-(CR) 35
12.1.1. Staad Output For Column CR 36
12.2. Design of Column -(C1) 37
12.2.1. Staad Output For Column C1 38
12.3. Design of Column -(C2) 39
12.3.1. Staad Output For Column C2 40
12.4. Design of Column -(C3) 41
12.4.1. Staad Output For Column C3 42
13. CANTILEVER SLAB DESIGN 43
14. FOUNDATION DESIGN 45
References 46
ATTACHMENT 1 STAAD INPUT DATA 47
ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS 57

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of Laboratory Building in Process Plant Area.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Laboratory Building only.

PROPRIETARY & CONFIDENTIAL

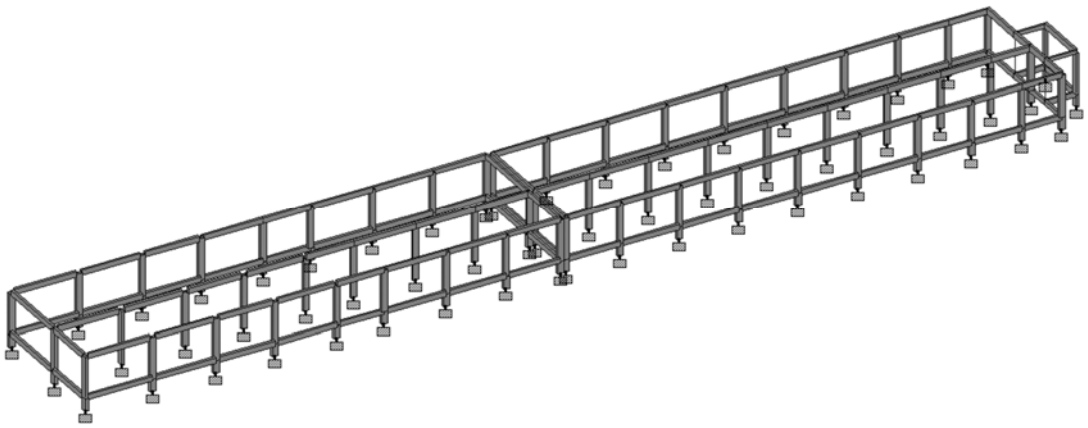
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HCS (Hollow Core Slab)

The Superstructure for this Accommodation Building was modeled using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & also keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base which are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.



STAAD 3D MODEL

4.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel

5. DESIGN LOADS

5.1. Dead Load

5.1.1. Roof Dead Loads

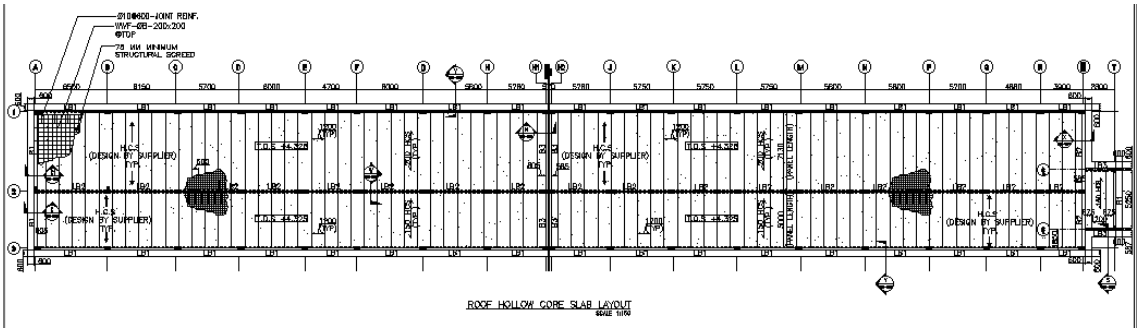
- Provision for Electrical Lighting = 0.05 kN/m²

Ceiling

Suspended steel channel system = 0.10 kN/m²

Acoustical Fiber Tile (16mm) = 0.05 kN/m²

Sub-total = 0.2 kN/m²



- Hollow Core Slab (PCI 6th Edition)

(HCS-200 Topped) b/w Grid 2 & 3 = **3.10 KN/m2**

(HCS-150 Topped) b/w Grid 1 & 2, Grid S&T = **2.52 KN/m2**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Prestressed Hollowcore Floor										
Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m^2 for Finishes Characteristic Service Loads Kn/m^2							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/m²**

Total Load b/w Grid 2 & 3 = **6.20 kN/m²**

Total Load b/w Grid 1 & 2, Grid S&T = **5.82 kN/m²**

UDL on External Beams along Grid Line-1 = 7.190 x 6.20/2 =**23 Kn/m**

UDL on External Beams along Grid Line-3 = 5.160 x 5.82/2 =**15.00 Kn/m**

UDL on Internal Beams along Grid Line-2 = 23+15 =**38.00 Kn/m**

UDL on External Beams along Grid Line-S & T = 5.240 x 5.82/2 =**15.25 Kn/m**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+ (0.05+0.2)/2x20)*0.6*0.6/2 = **1 knm/m**

5.1.2. Ground Floor Walls

- 250 thk. Outer CMU

16 x 0.25 x 3.5m clear height = **14.0 kN/m**

5.2. Live Load

5.2.1. Roof Live Loads

Live loads were taken from SEI/ASCE 7-05, Table 4-1.

Ordinary Flat = **0.50 kN/m²**

UDL on External Beams along Grid Line-1 = 7.190 x 0.50/2 =**1.80 Kn/m**

UDL on External Beams along Grid Line-3 = 5.160 x 0.50/2 =**1.29 Kn/m**

UDL on Internal Beams along Grid Line-2 = 1.80+1.29 =**3.00 Kn/m**

UDL on External Beams along Grid Line-S & T = 5.240 x 0.50/2 =**1.31 Kn/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 1*0.6*0.6/2 = **0.18 kN/m /m**

5.3. Wind Load

Wind loads were automatically generated in STAAD as per ASCE 7-05. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

PROPRIETARY & CONFIDENTIAL

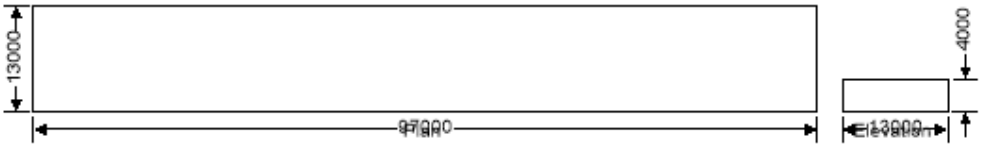
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

5.4. Wind Load (Analytical Verification)

WIND LOADING (SBC-301)

Using the directional design method



Building data

Type of roof;	Flat
Length of building;	b = 97000 mm
Width of building;	d = 13000mm
Height to eaves;	H = 4000 mm
Mean height;	h = 4000 mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff;	K_d = 0.85(As per SBC-301, Table 6.4-1)
Exposure category ;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification ;	Partially enclosed buildings(As per SBC-301, Section-6.2)
Internal pressure coef +ve;	GC_{pi_p} = 0.55(As per SBC-301, Figure-7.2-1)
Internal pressure coef –ve;	GC_{pi_n} = -0.55(As per SBC-301, Figure-7.2-1)
Gust effect factor;	G_f = 0.85(As per SBC-301, Section-7.2.7.1)
Importance Factor;	1(As per SBC-301, Section-9.5)

Topography

Topography factor not significant;	K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)
Velocity pressure equation;	$q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2 \times I$

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m^2)
3900	1.03	1.29
4000	1.03	1.29

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	4.00	0.80	1.29	0.17	387.90	64.98
B	4.00	-0.50	1.29	-1.26	388.00	-487.47
C	4.00	-0.70	1.29	-1.48	52.00	-76.72
D	4.00	-0.70	1.29	-1.48	52.00	-76.72

Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	4.00	0.80	1.29	1.58	387.90	614.80
B	4.00	-0.50	1.29	0.16	388.00	62.50
C	4.00	-0.70	1.29	-0.06	52.00	-3.02
D	4.00	-0.70	1.29	-0.06	52.00	-3.02

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

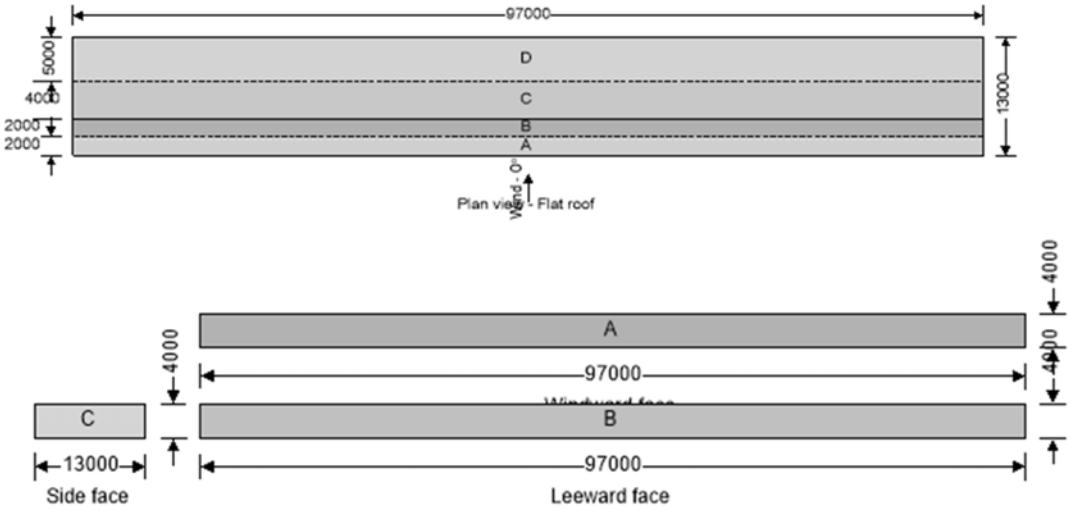
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	4.00	0.80	1.29	0.17	51.99	8.71
B	4.00	-0.20	1.29	-0.93	52.00	-48.24
C	4.00	-0.70	1.29	-1.48	388.00	-572.46
D	4.00	-0.70	1.29	-1.48	388.00	-572.46

Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	4.00	0.80	1.29	1.58	51.99	82.40
B	4.00	-0.20	1.29	0.49	52.00	25.46
C	4.00	-0.70	1.29	-0.06	388.00	-22.50
D	4.00	-0.70	1.29	-0.06	388.00	-22.50



So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

5.5. Seismic Load

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

S_{DS} = 2/3*Fa*S1 = 2/3 * 1.60 * 0.10 = 0.10 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration@1 second

S_{DS} Damped Design Spectral Response Acceleration@short period

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

5.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

7. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

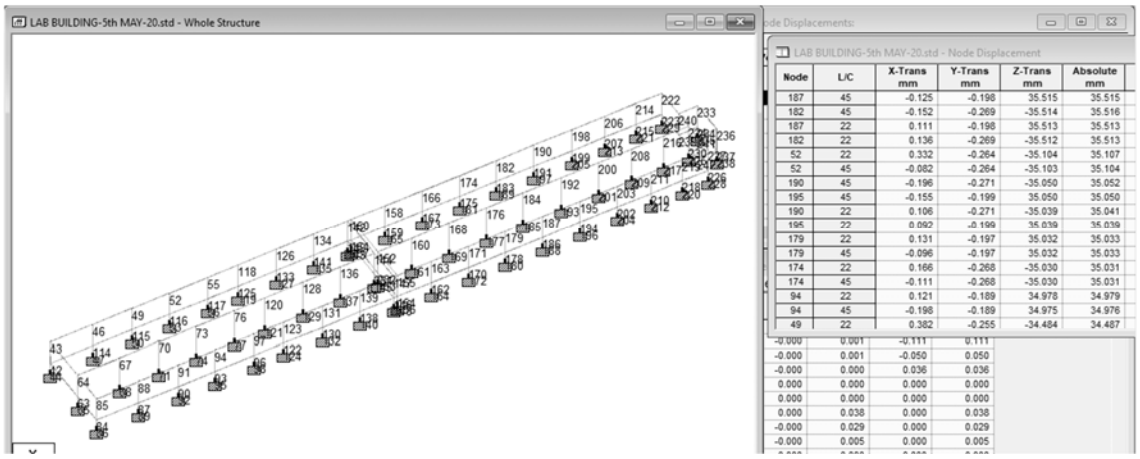
8. DEFLECTIONS/DRIFT CHECK

8.1. Maximum Allowable Deflection

L/240: for member load with live load plus dead load

L/600: for members supporting masonry under total loads

Where, L = span of member between support



Node	L/C	X-Trans mm	Y-Trans mm	Z-Trans mm	Absolute mm
187	45	-0.125	-0.198	35.515	35.515
182	45	-0.152	-0.269	-35.514	35.516
187	22	0.111	-0.198	35.513	35.513
182	22	0.136	-0.269	-35.512	35.513
52	22	0.332	-0.264	-35.104	35.107
52	45	-0.082	-0.264	-35.103	35.104
190	45	-0.196	-0.271	-35.050	35.052
195	45	-0.155	-0.199	35.050	35.050
190	22	0.106	-0.271	-35.039	35.041
195	22	0.092	-0.199	35.039	35.039
179	22	0.131	-0.197	35.032	35.033
179	45	-0.096	-0.197	35.032	35.033
174	22	0.166	-0.268	-35.030	35.031
174	45	-0.111	-0.268	-35.030	35.031
94	22	0.121	-0.189	34.978	34.979
94	45	-0.198	-0.189	34.975	34.976
49	22	0.382	-0.255	-34.484	34.487
-0.000	0.001	-0.111	0.111		
-0.000	0.001	-0.050	0.050		
-0.000	0.000	0.036	0.036		
0.000	0.000	0.000	0.000		
0.000	0.000	0.000	0.000		
0.000	0.038	0.000	0.038		
-0.000	0.029	0.000	0.029		
-0.000	0.005	0.000	0.005		

Maximum attained Drift as per Analysis from above figure = 35 mm

Allowable Drift as per Table-10.12, SBC-301-2007 is H/100 = 4000/100 = 40mm

Hence the Structure is safe considering drift check…

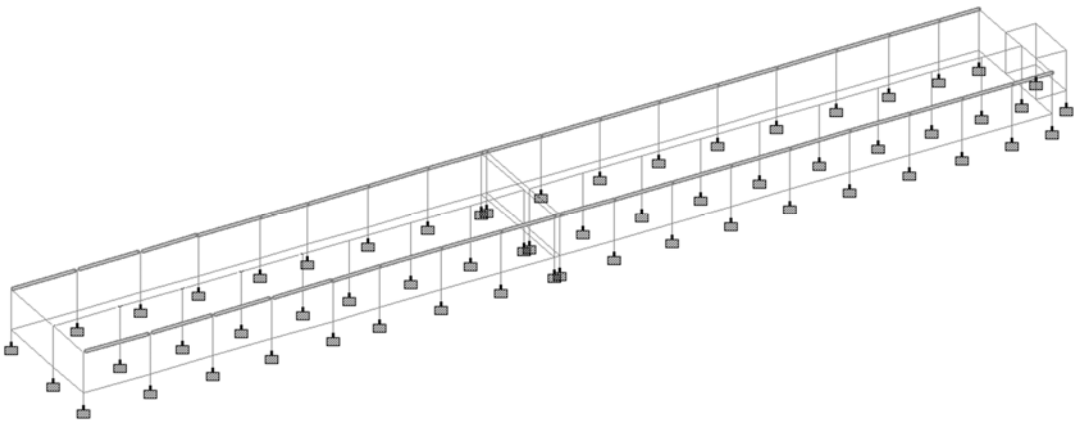
PROPRIETARY & CONFIDENTIAL

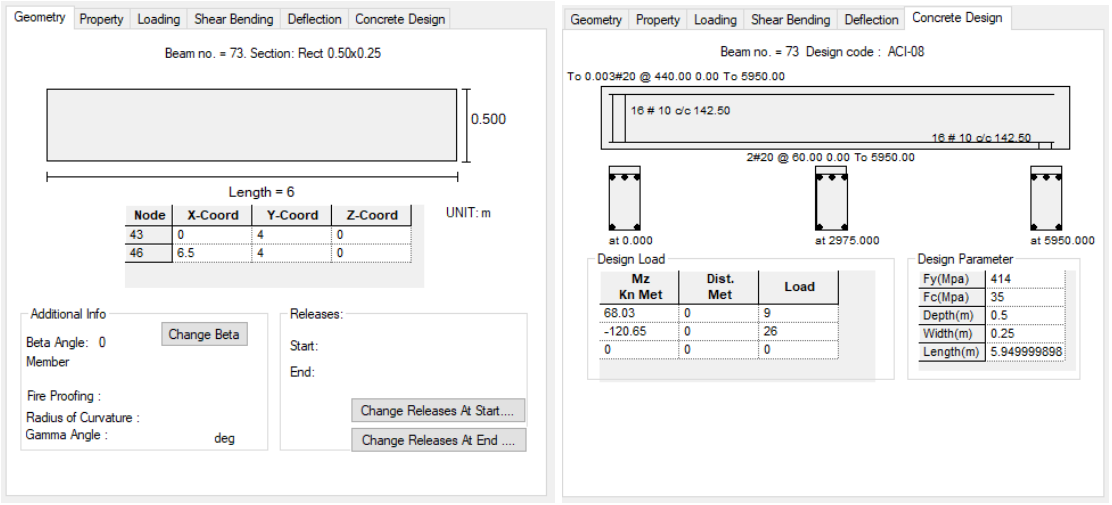
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9. CONCRETE ROOF BEAM DESIGN

9.1. Design of Ledger Beam(LB1)





Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = **120 Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Ast1.req = **830 mm²**

Ast.min= (As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **471mm²**

Maximum Moment required from Analysis = **63 Knm (Sagging)**

Ast2.req = **410 mm²**

However Provide 3-20 dia -Full Straight Bars Top & 3-20 dia-Full Straight Bottom.

Ast1.pro**>>>>>>(Ast.min, Ast1.req)**

Ast2.pro**>>>>>>(Ast.min, Ast2.req)**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

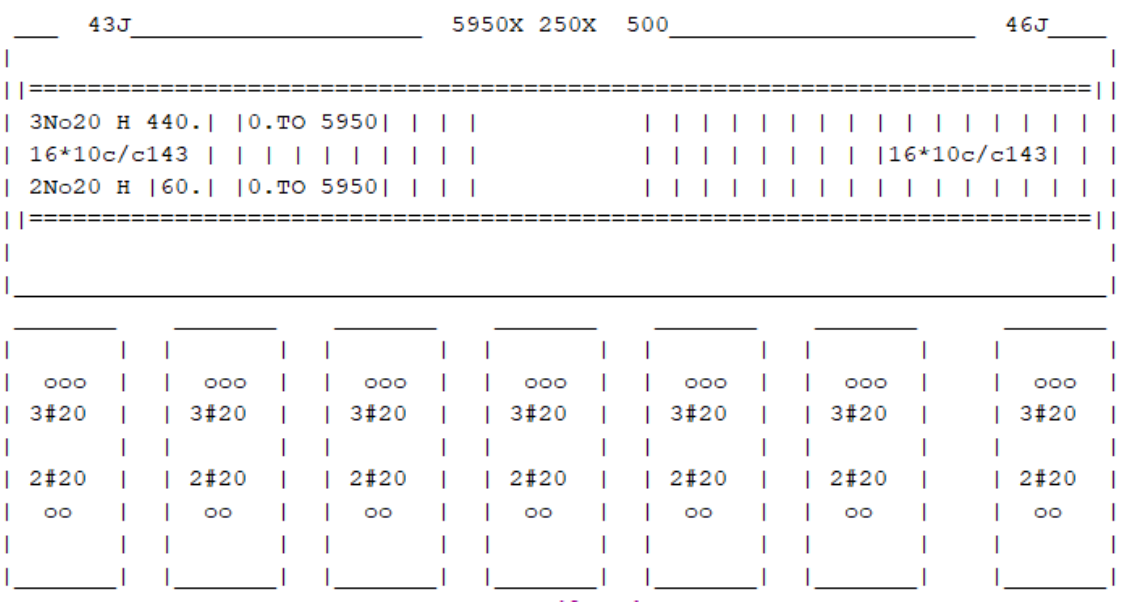
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.1.1. Staad Output For LB1

```
        B E A M  N O.    73 D E S I G N  R E S U L T S - SHEAR


  AT START SUPPORT - Vu=   59.45 KNS  Vc=  200.11 KNS  Vs=    0.00 KNS
  Tu=   16.44 KN-MET  Tc=    4.2 KN-MET  Ts=   21.9 KN-MET  LOAD    29
  STIRRUPS ARE REQUIRED FOR TORSION.
  REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
  PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2035. MM
  ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   5.42 SQ.CM.

  AT END   SUPPORT - Vu=   70.16 KNS  Vc=  114.34 KNS  Vs=    0.00 KNS
  Tu=   13.80 KN-MET  Tc=    4.0 KN-MET  Ts=   18.4 KN-MET  LOAD    28
  STIRRUPS ARE REQUIRED FOR TORSION.
  REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
  PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2035. MM
  ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   4.55 SQ.CM.
```

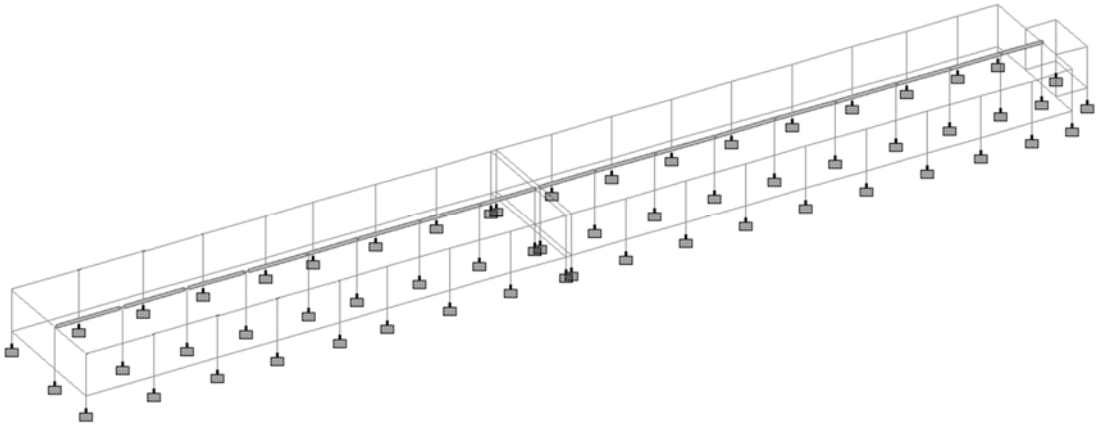


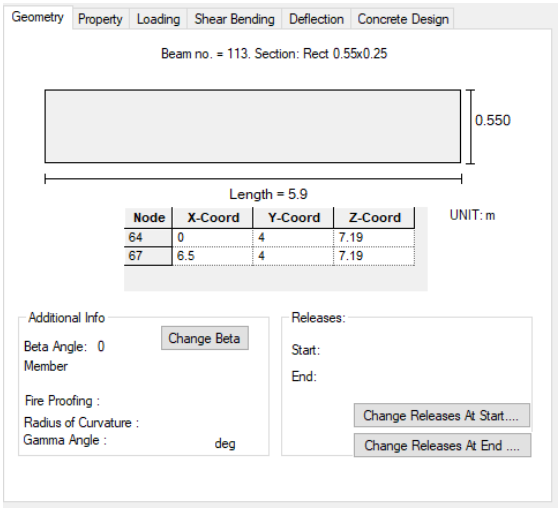
PROPRIETARY & CONFIDENTIAL

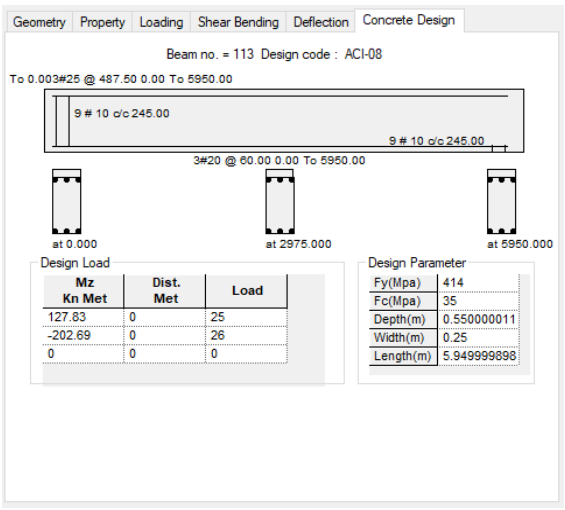
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.2. Design of Ledger Beam(LB2)







Design of Ledger Beam (LB2)

Provide 3-25 dia-Full Straight Bars Top & 3-25 dia-Full Straight Bottom.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

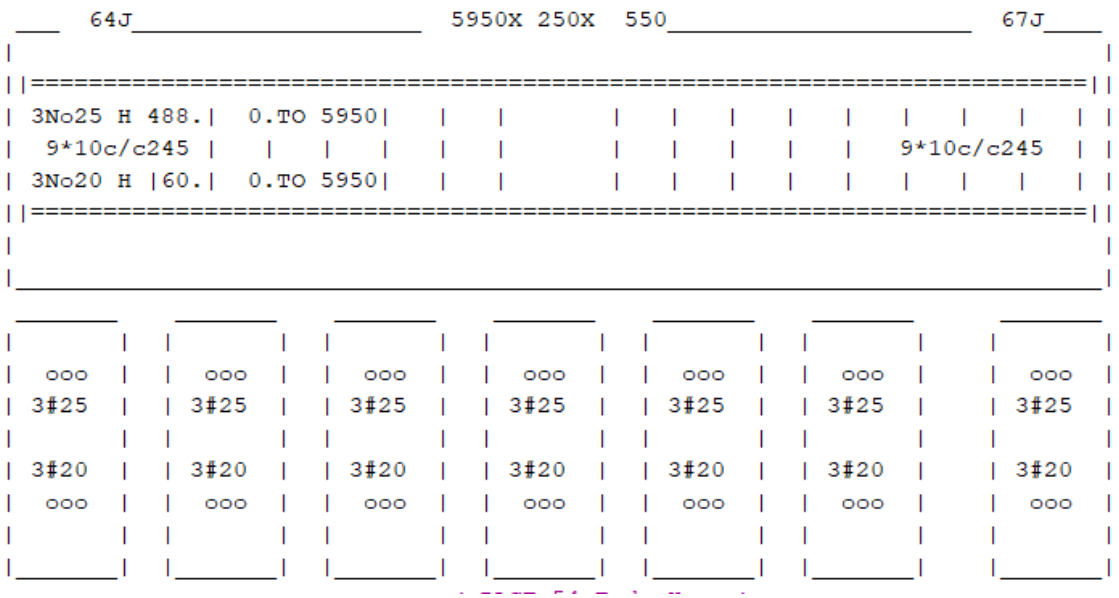
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.2.1. Staad Output For LB2

```
B E A M  N O.   113 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=    64.60 KNS   Vc=  126.16 KNS   Vs=     0.00 KNS
Tu=     0.06 KN-MET   Tc=     4.5 KN-MET   Ts=     0.0 KN-MET   LOAD      3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 245. MM  C/C FOR 1935. MM

AT END    SUPPORT - Vu=    95.00 KNS   Vc=  151.25 KNS   Vs=     0.00 KNS
Tu=     0.06 KN-MET   Tc=     4.5 KN-MET   Ts=     0.0 KN-MET   LOAD      3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 245. MM  C/C FOR 1935. MM
```

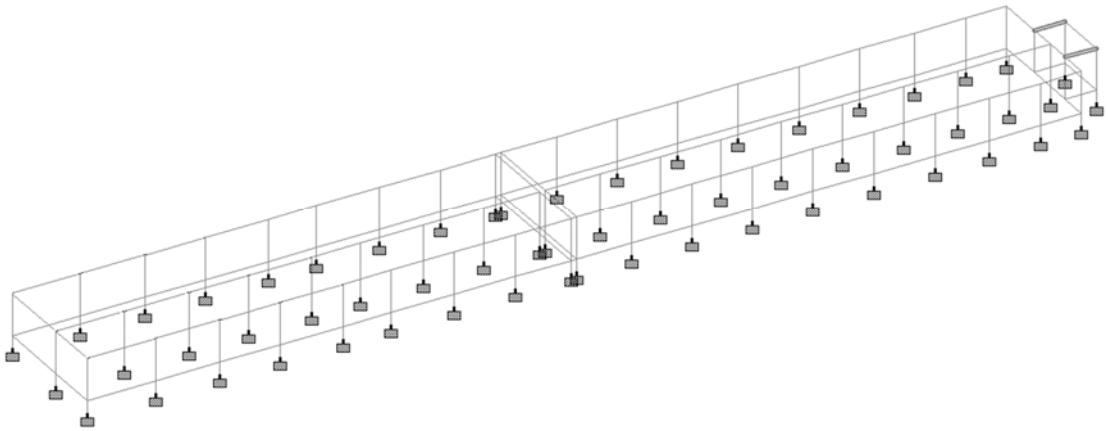


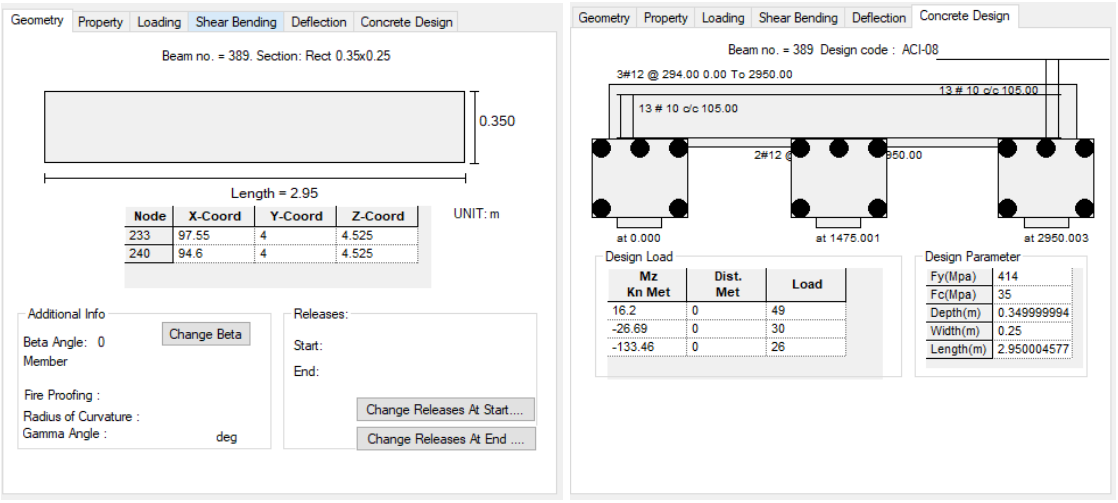
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.3. Design of Ledger Beam(LB3)





Design of Ledger Beam (LB3)

Provide 3-12 dia-Full Straight Bars Top & 3-12 dia-Full Straight Bottom.

PROPRIETARY & CONFIDENTIAL

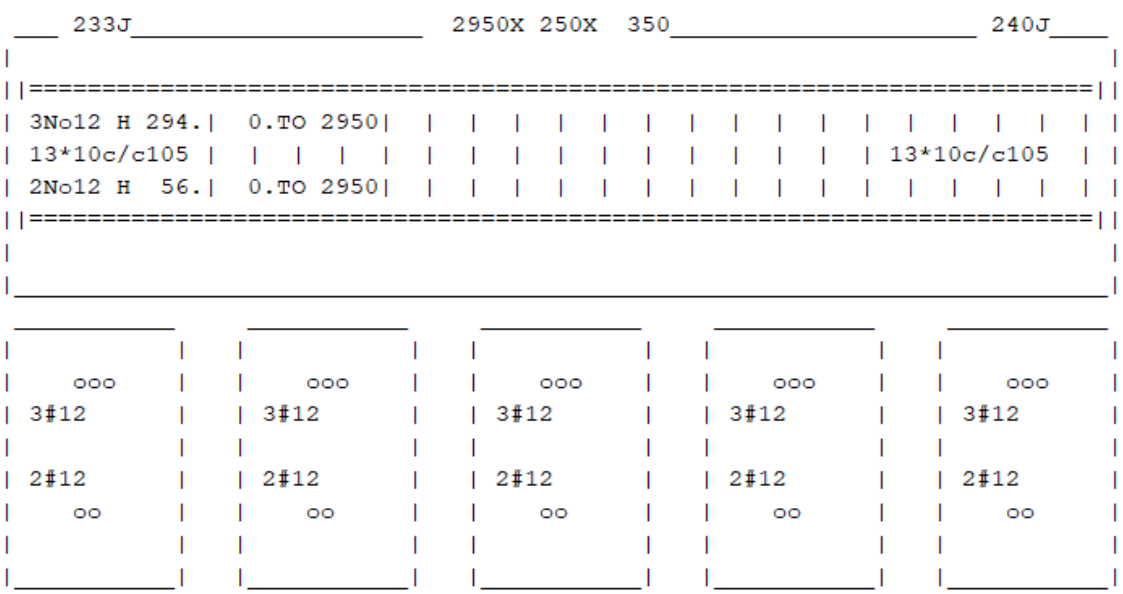
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.3.1. Staad Output For LB3

```
AT START SUPPORT - Vu=     4.35 KNS  Vc=    71.38 KNS  Vs=     0.00 KNS
Tu=     2.65 KN-MET  Tc=     2.3 KN-MET  Ts=     3.5 KN-MET  LOAD      9
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 105. MM  C/C FOR 1185. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.01 SQ.CM.

AT END   SUPPORT - Vu=     4.35 KNS  Vc=    71.38 KNS  Vs=     0.00 KNS
Tu=     2.65 KN-MET  Tc=     2.3 KN-MET  Ts=     3.5 KN-MET  LOAD      9
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 105. MM  C/C FOR 1185. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.01 SQ.CM.
```

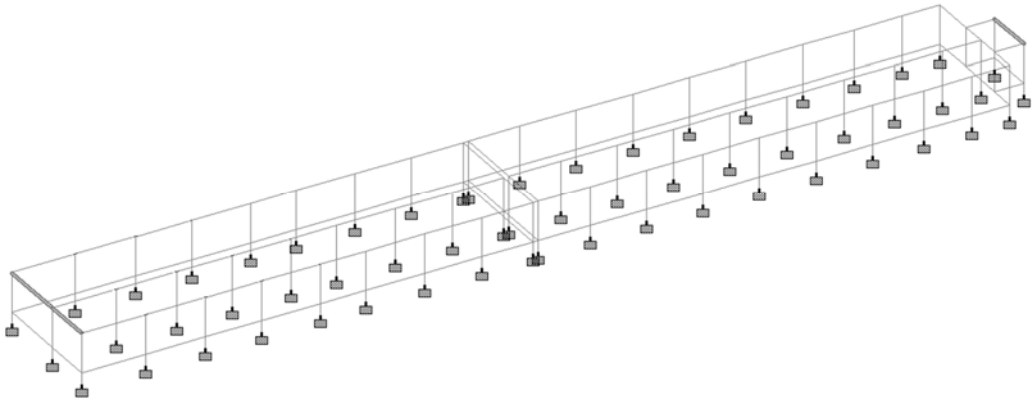


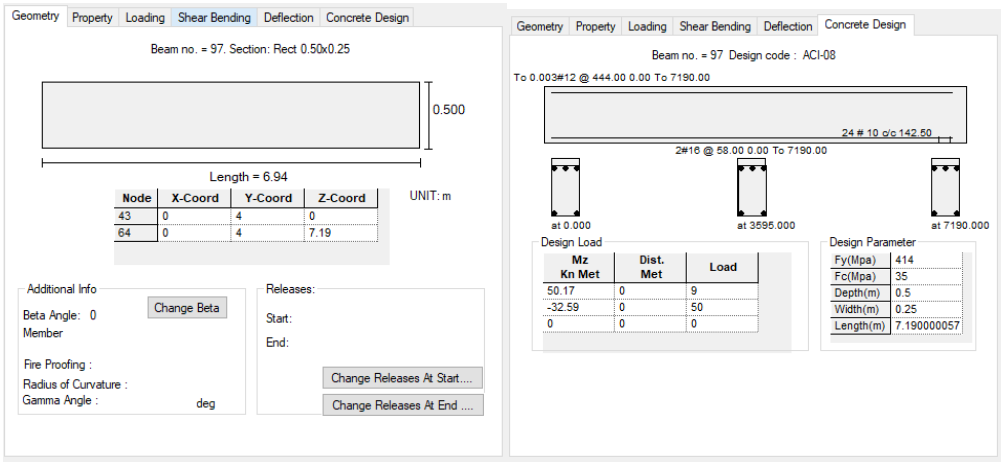
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.4. Design of Beam-(B1)





Design of Beam (B1)

However Provide minimum 3-16 dia-Full Straight Bars Top & Bottom.

PROPRIETARY & CONFIDENTIAL

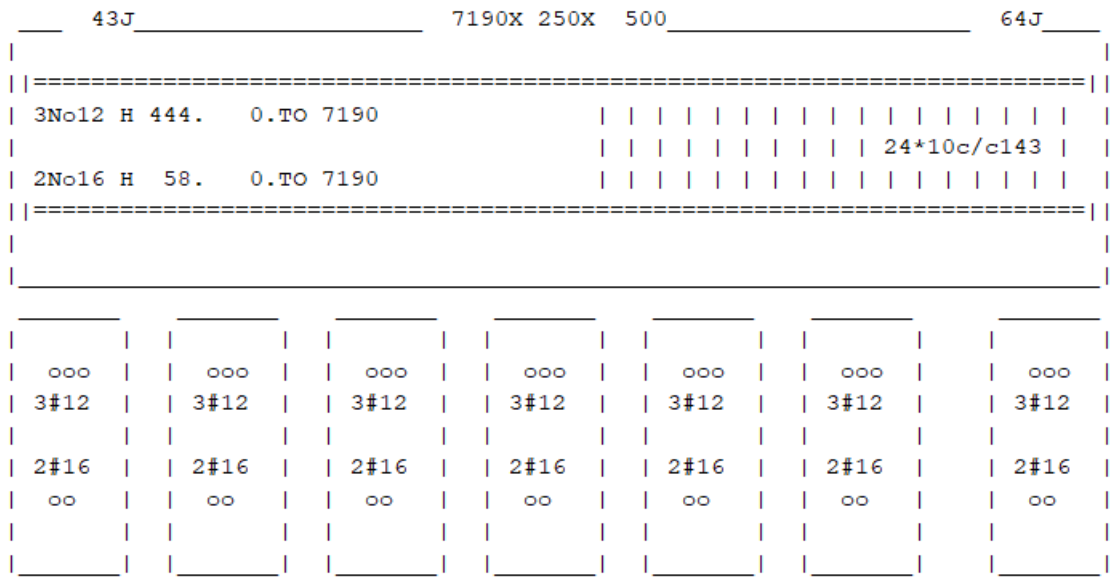
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.4.1. Staad Output For B1

```
        B E A M  N O.    97 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=     9.36 KNS   Vc=  106.99 KNS   Vs=     0.00 KNS
Tu=     0.60 KN-MET   Tc=     3.9 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=    12.88 KNS   Vc=  107.18 KNS   Vs=     0.00 KNS
Tu=     8.37 KN-MET   Tc=     3.9 KN-MET   Ts=    11.2 KN-MET   LOAD     25
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 3155. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    2.76 SQ.CM.
```

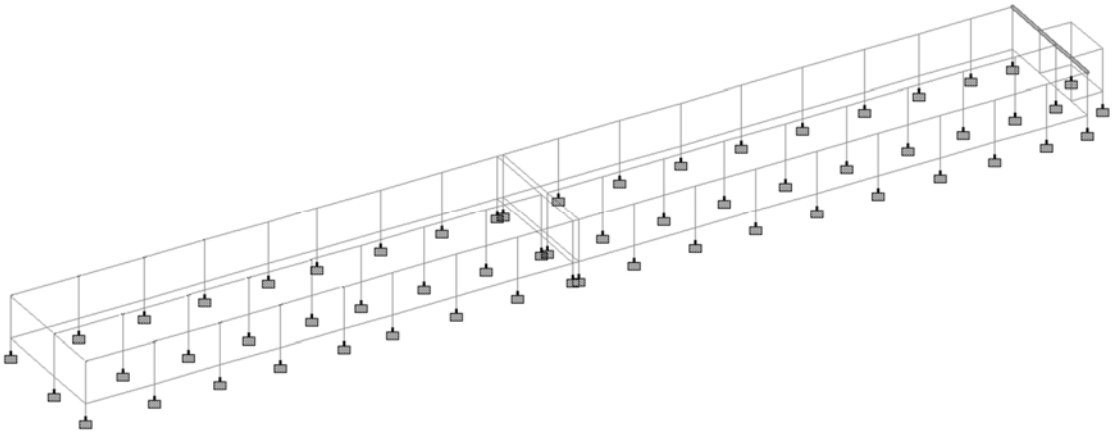


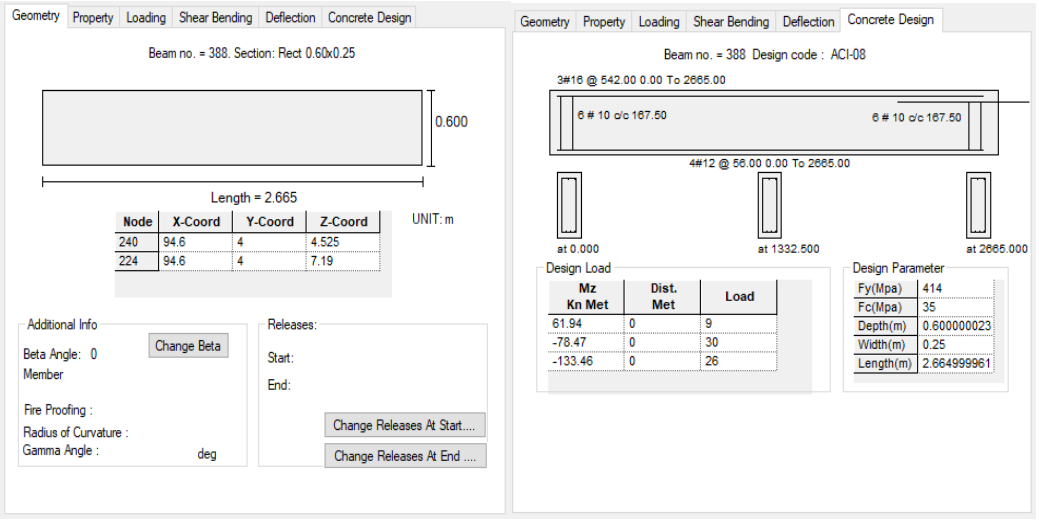
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.5. Design of Beam-(B2)





However Provide minimum 3-20 dia-Full Straight Bars Top & Bottom.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

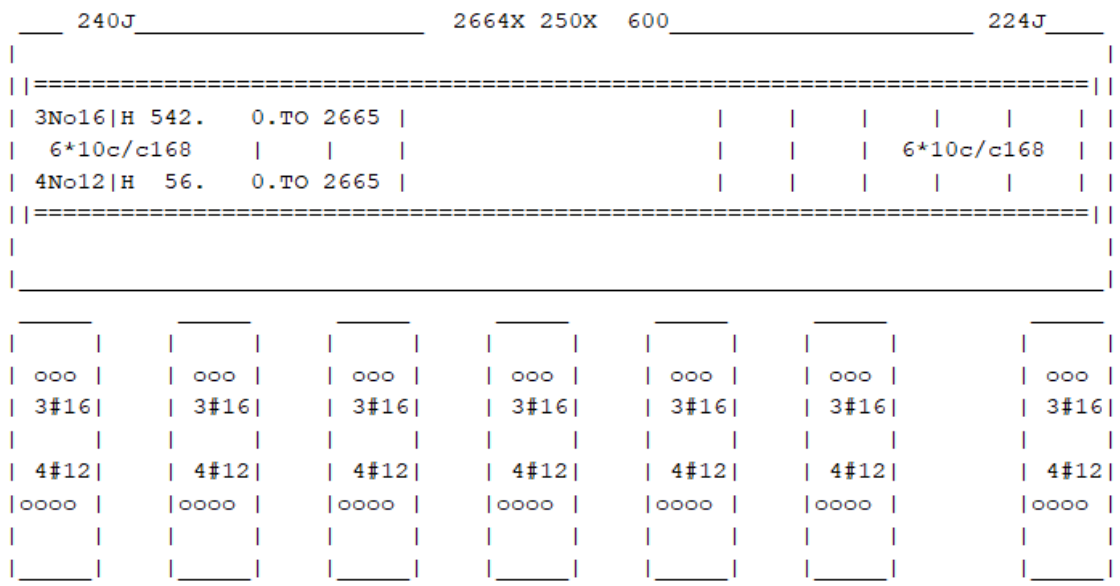
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.5.1. Staad Output For B2

```
        B E A M  N O.   388 D E S I G N  R E S U L T S - SHEAR


 AT START SUPPORT - Vu=     6.73 KNS   Vc=  131.70 KNS   Vs=     0.00 KNS
 Tu=     5.13 KN-MET   Tc=     4.8 KN-MET   Ts=     6.8 KN-MET   LOAD      9
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 168. MM  C/C FOR  792. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    1.60 SQ.CM.

 AT END    SUPPORT - Vu=     6.73 KNS   Vc=  131.70 KNS   Vs=     0.00 KNS
 Tu=     5.13 KN-MET   Tc=     4.8 KN-MET   Ts=     6.8 KN-MET   LOAD      9
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 168. MM  C/C FOR  792. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    1.60 SQ.CM.
```

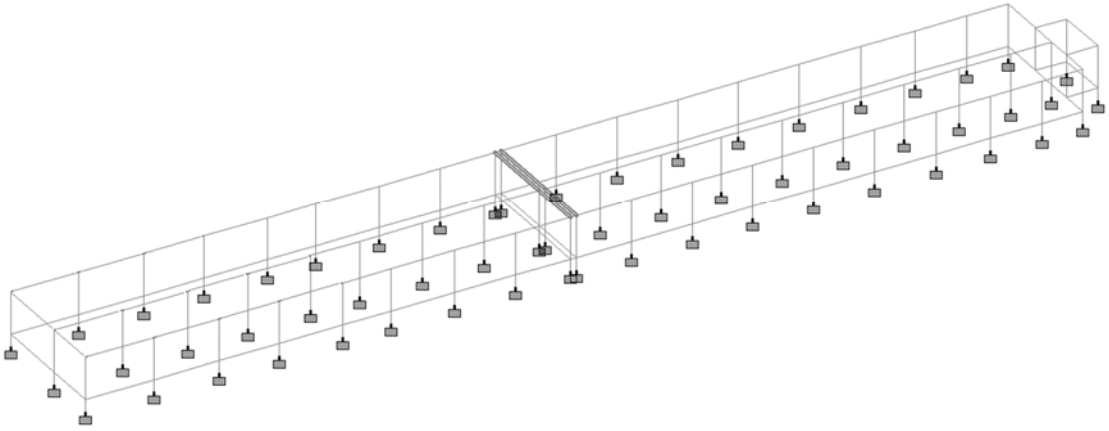


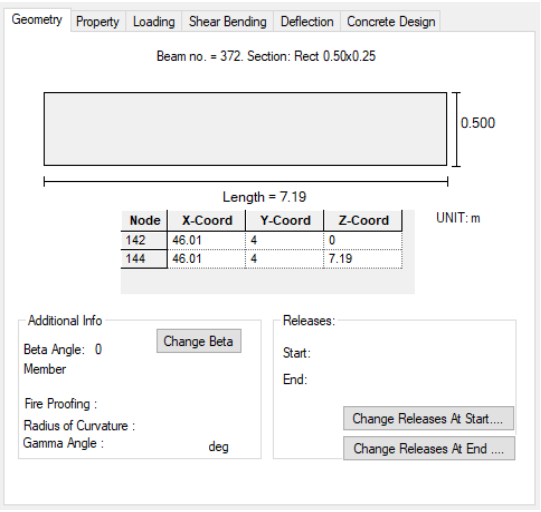
PROPRIETARY & CONFIDENTIAL

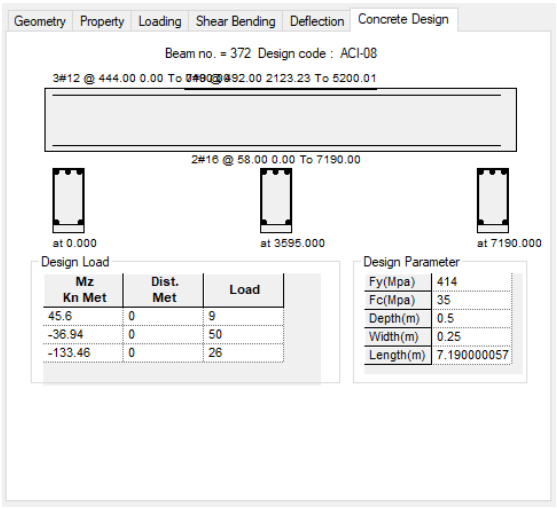
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.6. Design of Beam-(B3)







However Provide minimum 3-16 dia-Full Straight Bars Top & Bottom.

PROPRIETARY & CONFIDENTIAL

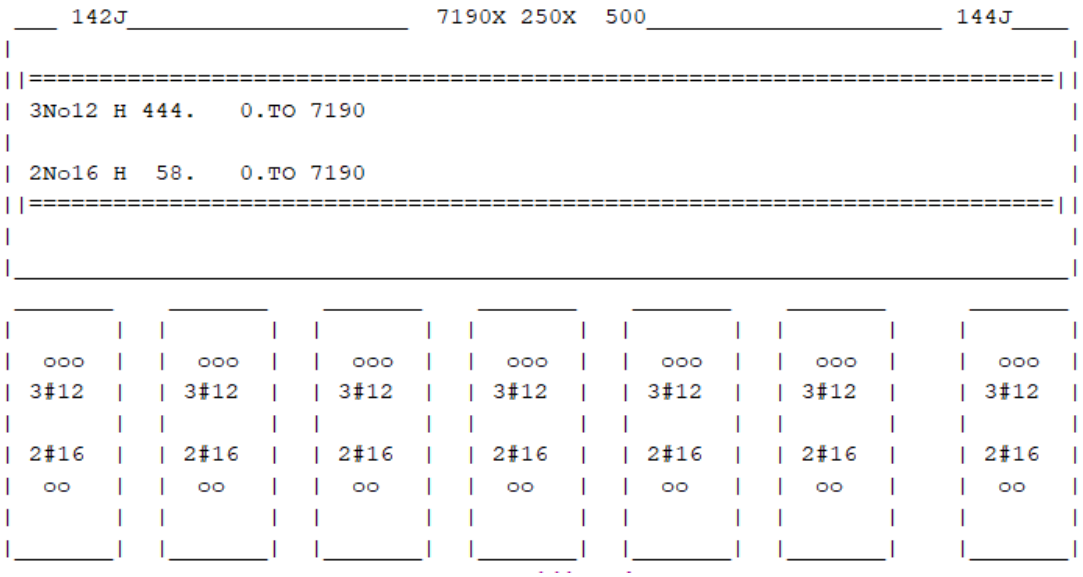
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.6.1. Staad Output For B3

```
                B E A M   N O.   372 D E S I G N   R E S U L T S - SHEAR


   AT START SUPPORT - Vu=     8.95 KNS   Vc=  107.81 KNS   Vs=     0.00 KNS
   Tu=     0.65 KN-MET   Tc=     3.9 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                      STIRRUPS ARE NOT REQUIRED.
   AT END   SUPPORT - Vu=     9.63 KNS   Vc=  106.75 KNS   Vs=     0.00 KNS
   Tu=     0.65 KN-MET   Tc=     3.9 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                      STIRRUPS ARE NOT REQUIRED.
```



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

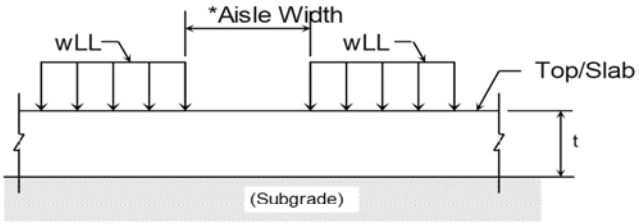
10. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm	
Concrete Strength, f 'c =	5000	psi	35.000	kpa	
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)
Factor of Safety, FS =	1.500		1.500		
Uniform Live Load, wLL =	104.45	psf	5.00	kpa	



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Allow. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

Provide WWF-6mm Dia@200 c-c Both Ways-TOP

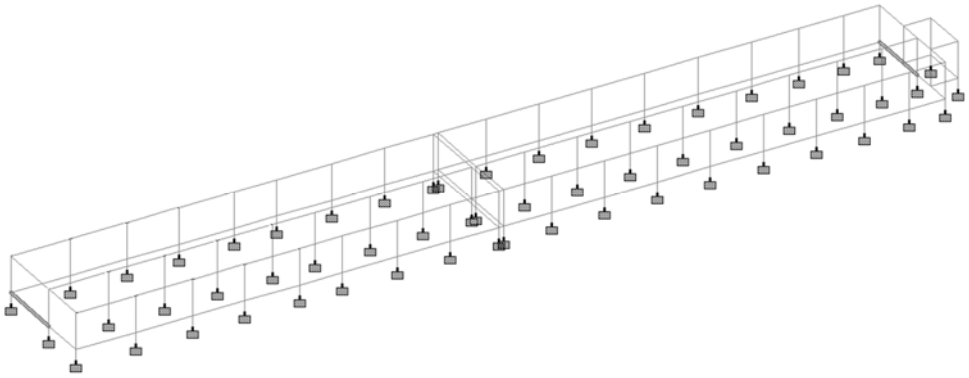
PROPRIETARY & CONFIDENTIAL

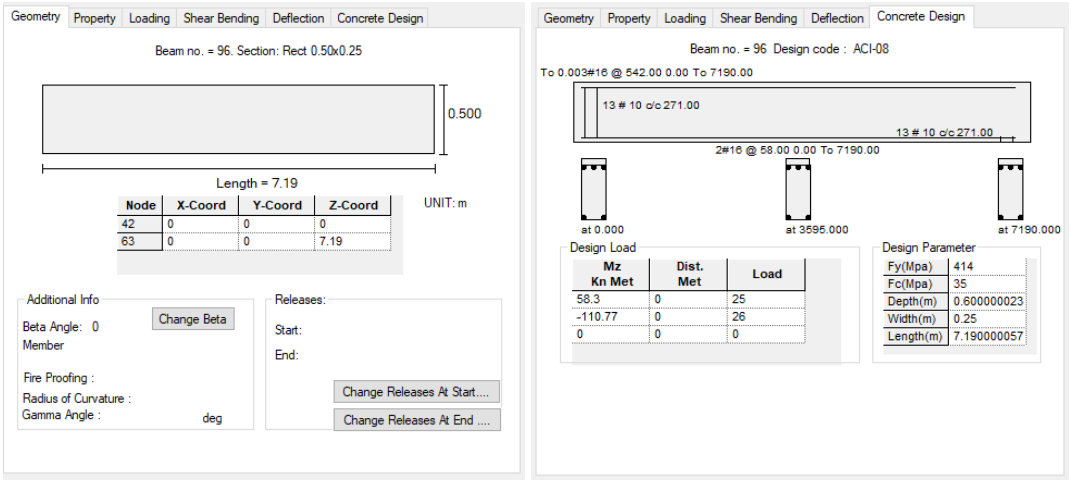
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

11. DESIGN OF GRADE BEAM-(GB1)

11.1. Design of Beam-(GB1)





Provide minimum 3-20 dia-Full Straight Bars Top & Bottom.

PROPRIETARY & CONFIDENTIAL

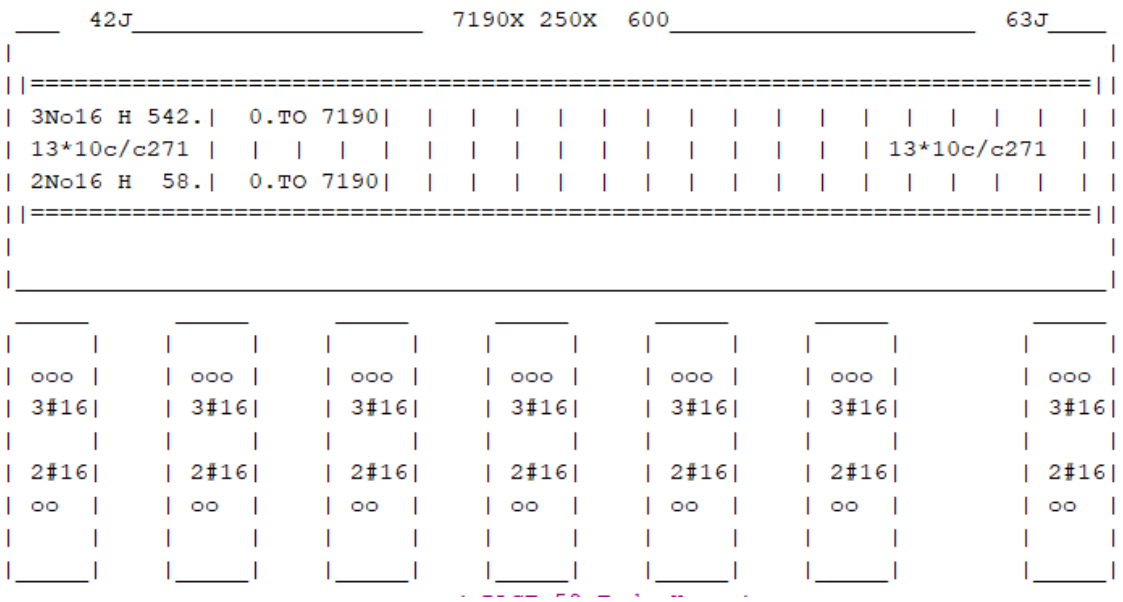
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

11.1.1. Staad Output for GB1

```
AT START SUPPORT - Vu=   51.50 KNS  Vc=  134.61 KNS  Vs=    0.00 KNS
Tu=    1.18 KN-MET  Tc=    5.3 KN-MET  Ts=    0.0 KN-MET  LOAD     3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 271. MM  C/C FOR 3055. MM

AT END   SUPPORT - Vu=   55.64 KNS  Vc=  132.15 KNS  Vs=    0.00 KNS
Tu=    1.18 KN-MET  Tc=    5.3 KN-MET  Ts=    0.0 KN-MET  LOAD     3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 271. MM  C/C FOR 3055. MM
```

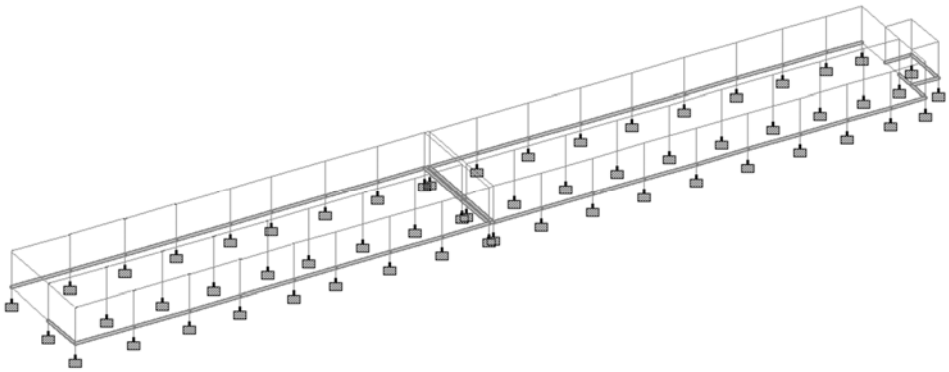


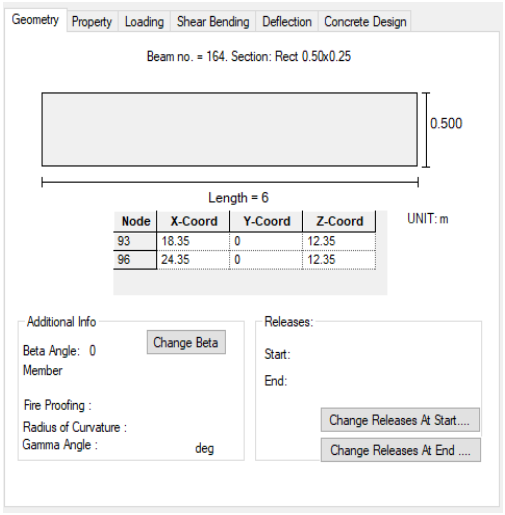
PROPRIETARY & CONFIDENTIAL

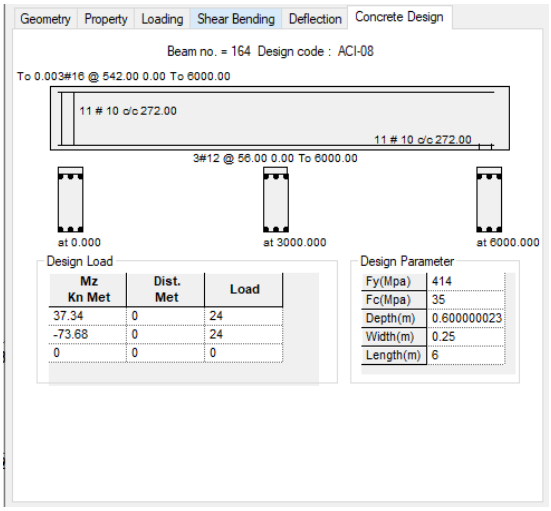
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

11.2. Design of Beam-(GB2)







Provide minimum 3-16 dia-Full Straight Bars Top & Bottom.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

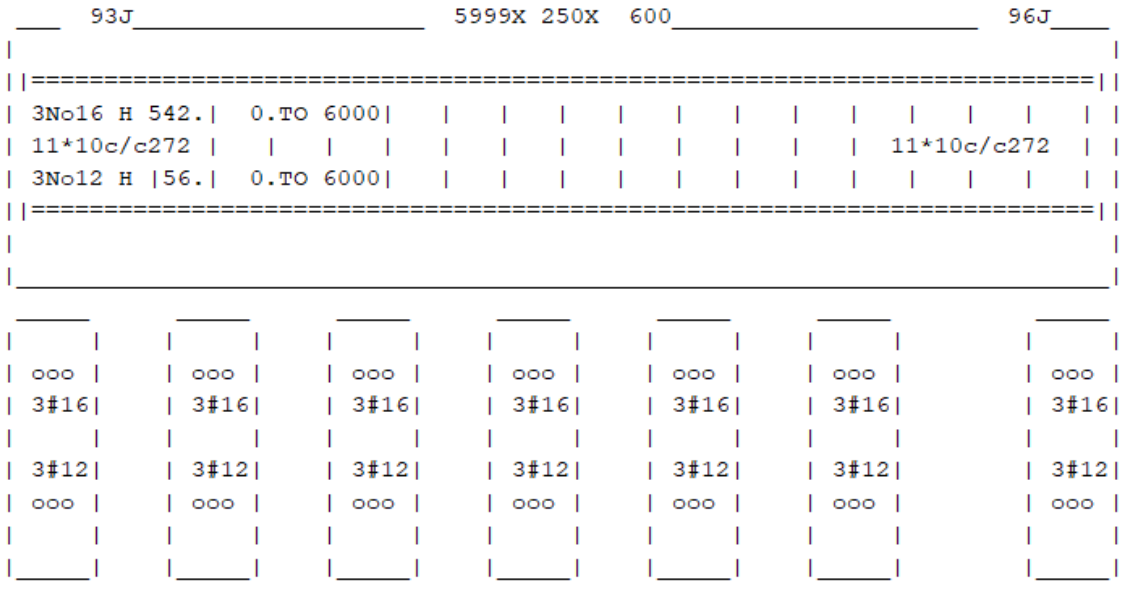
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

11.2.1. Staad Output for GB2

```
                B E A M  N O.   164 D E S I G N  R E S U L T S - SHEAR


   AT START SUPPORT - Vu=   60.57 KNS  Vc=  130.78 KNS  Vs=    0.00 KNS
   Tu=    0.01 KN-MET  Tc=    4.9 KN-MET  Ts=    0.0 KN-MET  LOAD    24
   NO STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 272. MM  C/C FOR 2460. MM

   AT END   SUPPORT - Vu=   60.21 KNS  Vc=  130.85 KNS  Vs=    0.00 KNS
   Tu=    0.01 KN-MET  Tc=    4.9 KN-MET  Ts=    0.0 KN-MET  LOAD    24
   NO STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 272. MM  C/C FOR 2460. MM
```



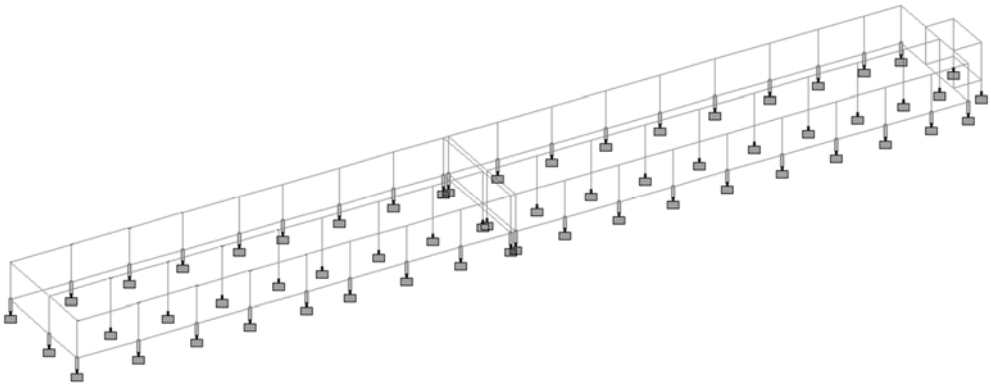
PROPRIETARY & CONFIDENTIAL

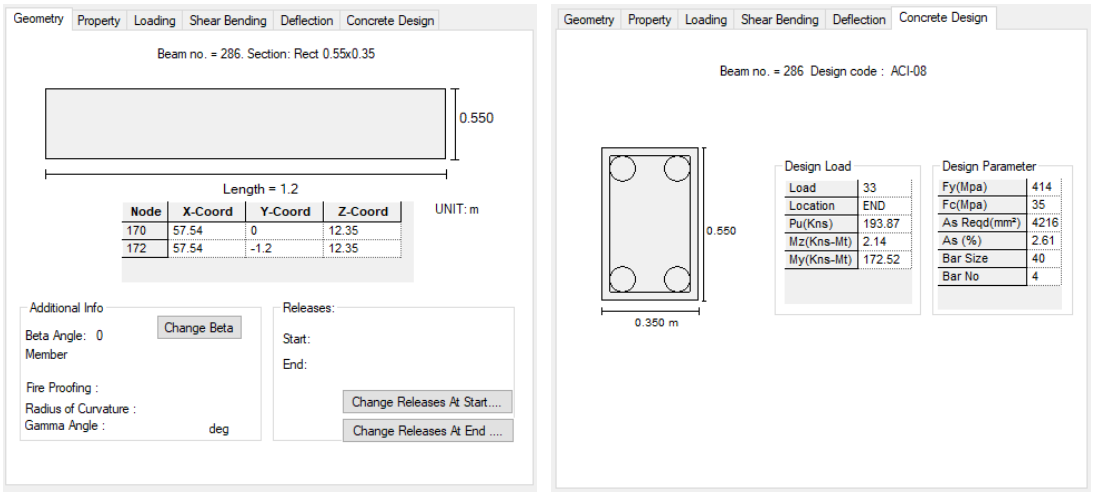
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

12. CONCRETE COLUMN DESIGN

12.1. Design of Column-(CR)





Column Design CR

Provide equivalent 10 No's-Día 25 equivalent Vertical Reinforcement in the Column Neck.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

12.1.1. Staad Output For Column CR

```
      COLUMN  NO.   286  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 350.0 X 550.0 MMS, TIED
        AREA OF STEEL REQUIRED =  4215.8  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

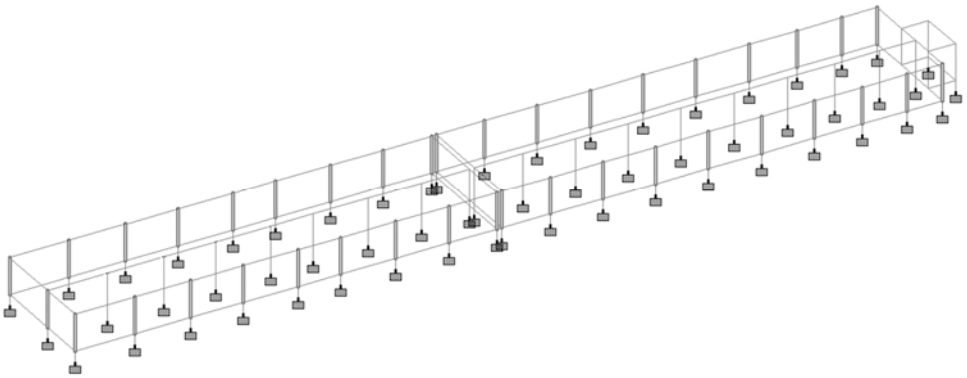
  4 - 40 MM               2.611       33      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 350.00 MM
```

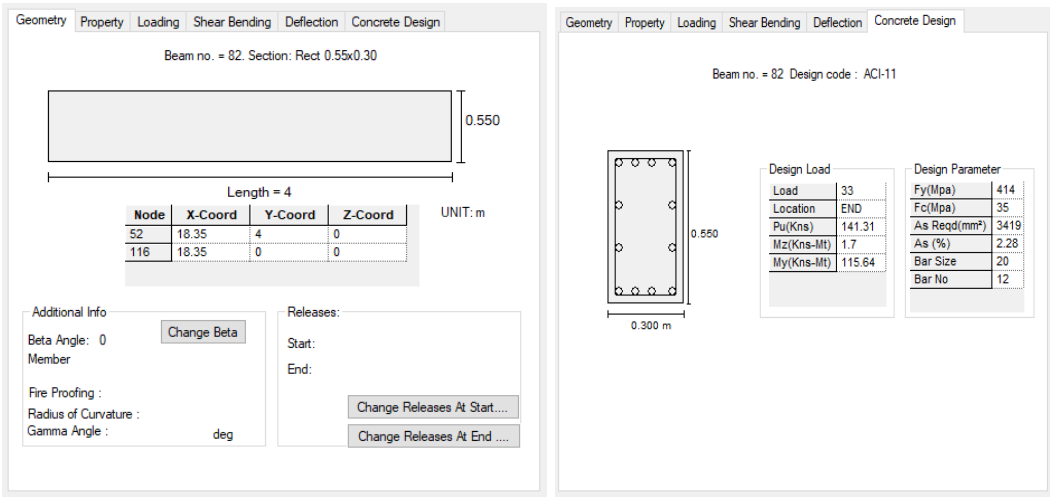

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

12.2. Design of Column -(C1)





Column Design C1

Provide equivalent 8 No's-Dia 25 equivalent Vertical Reinforcement in the Column C1.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

12.2.1. Staad Output For Column C1

```
      ACI 318-11   COLUMN  NO.    82   DESIGN RESULTS
      ================================================


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 300.0 X 550.0 MMS, TIED
        AREA OF STEEL REQUIRED =  3418.8  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

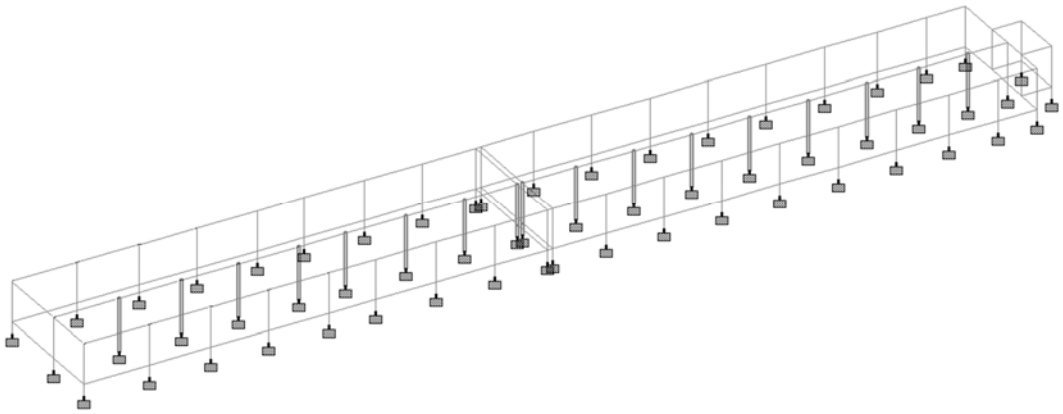
 12 - 20 MM               2.285       33      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 300.00 MM
```

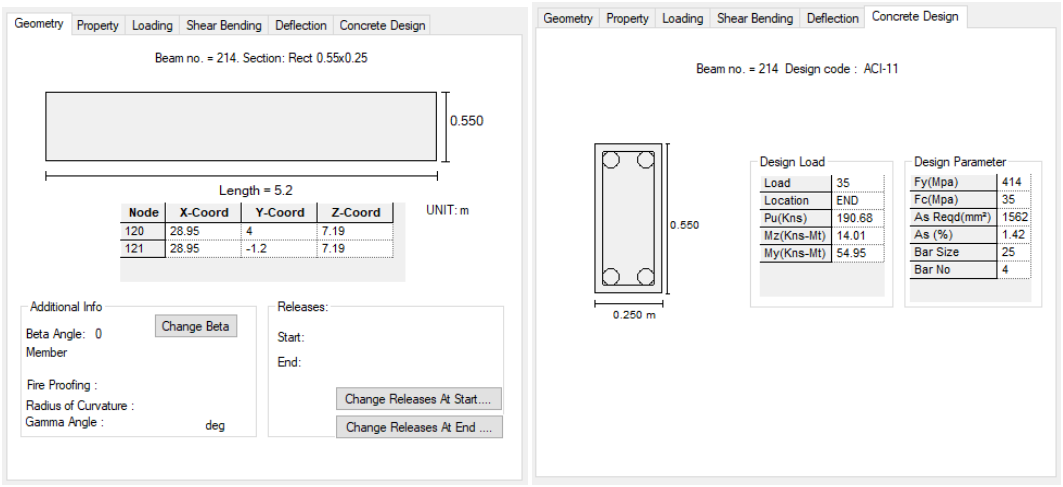

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

12.3. Design of Column -(C2)





Column Design C2

Provide equivalent 8 No's-Día 20 equivalent Vertical Reinforcement in the Column C2.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

12.3.1. Staad Output For Column C2

```
        ACI 318-11   COLUMN  NO.   214   DESIGN RESULTS
        ==================================================


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 250.0 X 550.0 MMS, TIED
        AREA OF STEEL REQUIRED =  1562.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

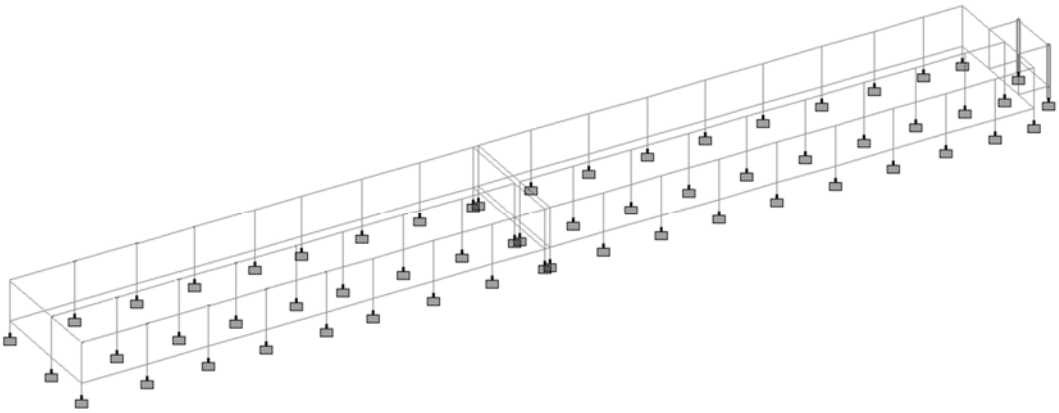
  4 - 25 MM               1.428       35      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 250.00 MM
```

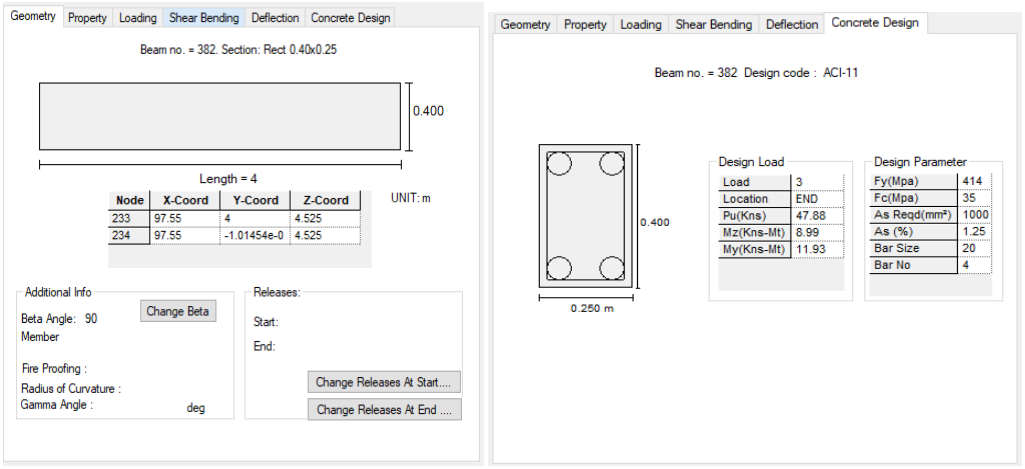

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

12.4. Design of Column -(C3)





Column Design C3

Provide minimum 4 No’s-Dia 20 Vertical Reinforcement in the Column C3.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

12.4.1. Staad Output For Column C3

```
      ACI 318-11   COLUMN  NO.   382   DESIGN RESULTS
      ================================================


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 250.0 X 400.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  1000.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

  4 - 20 MM               1.257        3      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 250.00 MM
```


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

13. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties			
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r	24		kN/m^3
Clear Span	0.6	met	Strength of concrete "f'_c"	35		N/mm^2
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420		N/mm^2
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20		mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9		
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75		
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es"	2E+05		N/mm^2
Loading			Modulus of Elasticity of concrete	27806		N/mm^2
Self Weight of slab	3.6	kN/m^2	Modular Ratio "n = Es/Ec"	7.19		
Floor Finish	3.5	kN/m^2	**Provided Reinforcement**			
Wall load	0	kN/m^2	Top layer Nos.	6	Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m^2	Bottom layer Nos.	0	Φ 12	mm
Live load	1	kN/m^2	Area of Top steel "$A_{s\ t}$"		468	mm^2
Total Ultimate Load "$Uu_{,tot}$"$_{(1.2D+1.6L)}$	10.12	kN/m^2	Area of Bottom steel "$A_{s\ b}$"		0	mm^2
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"d_b' = 10 mm			
Calculations						
Shear force Vu = U*le	**8.096**	**kN**				
Neg. Moment Mu = U*le^2/2	**3.51**	**kNm**				

Check for Section capacity								
Depth of comp. block "a" = $A_{s\ t}$*f_y/(0.85*f'_c*b)	6.61	mm						
Section capacity ΦMn = Φ*$A_{s\ t}$*f_y*(d-a/2)	**21.94**	**kNm**	**O.K**	>	**3.51**	**kNm**	(Mu)	
Shear requirements								
Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	>	**8.096**	**kN**	(Vu)	
Serviceability requirements								
d' = c+d_b/2	26	mm						
B = b/(n*$A_{s\ t}$)	0.297							
r = (n-1)*$A_{s\ b}$/(n*$A_{s\ t}$)	0							
k_d = (√(2*d*B(1+r*d'/d)+(1+r)^2)-(1+r))/B	29.01	mm						
I_{cr} = bkd^3/3+n*$A_{s\ t}$*(d-k_d)^2+(n-1)*$A_{s\ b}$*(k_d-d'	3.915E+07	mm^4						
Calculate Gross moment of inertia								
Area of slab A = t*b	2E+05	mm^2						
Depth of neutral axis "y"	75	mm						
Distance of top fiber from N.A "y_t"	75	mm						
Gross moment of inertia "I_g"	2.813E+08	mm^4						
Allowable stress fr = 0.62√f'c	3.67	mm^2						
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm						

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Service moments

Moment due to dead load "M_{dl}"	2.54	kNm	
Moment due to live load "Mll"	0.32	kNm	
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm	
Deflection due to dead + live loads			
R(dl+ll) = M_{cr}/Ms	1.000		
$I_e(dl+ll) = R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
$\delta = K*5/48*Ms*le^2/(Ec*I_e)$	0.08	mm	**K = 3.2 (conservative)**

Deflection due to dead load

R(dl) = M_{cr}/Md	5.411		
$I_e(dl) = R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4	
$\delta = K*5/48*Md*le^2/(Ec*I_e)$	0.00	mm	

Deflection due to live load

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load =
per ACI 318 Tab 9.5b $I_e/360$ **2.22 mm > 0.08 mm O.K**

Long term deflection

% of liveload sustained "SL"	50	%
$M_{sus} = M_{dl}+SL*M_{ll}$	2.70	kNm
R(sus) = $M_{cr}/Msus$	5.091	
$I_e(sus) = R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10	
$\delta = K*5/48*Msus*le^2/(Ec*I_e)$	0.00	mm

Magnification factor for creep and shrinkage
$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000
Where:-
ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318
$\rho' = A_{s\ b}/(b*d)$ 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

Allowable deflection = $I_e/240$ 3.333 mm > 0.079 mm O.K

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

14. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-RPT-1008 for Details)

TABLE 4: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Village Area	BH-21	Village	Isolated Square	1.20	1.0 x 1.0	1000	350	350	25	42,000	32,200
				1.50		1100	500	500	25	60,000	38,900
				1.20	2.0 x 2.0	700	580	580	25	69,600	37,500
				1.50		850	750	750	25	90,000	44,200
				1.20	3.0 x 3.0	600	750	600	25	72,000	42,800
				1.50		700	930	700	25	84,000	49,500
				1.20	4.0 x 4.0	550	910	550	25	66,000	48,100
				1.50		625	1090	625	25	75,000	54,800
				1.20	5.0 x 5.0	475	1060	475	25	57,000	53,400
				1.50		550	1240	550	25	66,000	60,200

Refer Attachment- II for Foundation Design calculations

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

REFERENCES

1.)	BUILDING PLANS	MD-216-49BL-EG-AR-GAD-3011
2.)	LAYOUT OF FDN, COL. & GRADE BEAM	MD-216-49BL-EG-CV-GAD-1063
3.)	LAYOUT OF ROOF & GRADE SLAB	MD-216-49BL-EG-CV-GAD-1064
4.)	DETAILS OF FDN, COL & GRADE BEAM	MD-216-49BL-EG-CV-DED-1065
5.)	DETAILS OF ROOF & GRADE SLAB	MD-216-49BL-EG-CV-DED-1066
6.)	GEOTECHNICAL REPORT	MD-216-3000-EG-CV-RPT-1008
7.)	STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001
8.)	BUILDING ELEVATIONS & SECTIONS	MD-216-49BL-EG-AR-GAD-3012

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

ATTACHMENT 1 STAAD INPUT DATA

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18-Oct-19
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
42 0 0 0; 43 0 4 0; 44 0 -1.2 0; 46 6.5 4 0; 47 6.5 -1.2 0; 49 12.65 4 0;
50 12.65 -1.2 0; 52 18.35 4 0; 53 18.35 -1.2 0; 55 24.35 4 0; 56 24.35 -1.2 0;
63 0 0 7.19; 64 0 4 7.19; 65 0 -1.2 7.19; 67 6.5 4 7.19; 68 6.5 -1.2 7.19;
70 12.65 4 7.19; 71 12.65 -1.2 7.19; 73 18.35 4 7.19; 74 18.35 -1.2 7.19;
76 24.35 4 7.19; 77 24.35 -1.2 7.19; 84 0 0 12.35; 85 0 4 12.35;
86 0 -1.2 12.35; 87 6.5 0 12.35; 88 6.5 4 12.35; 89 6.5 -1.2 12.35;
90 12.65 0 12.35; 91 12.65 4 12.35; 92 12.65 -1.2 12.35; 93 18.35 0 12.35;
94 18.35 4 12.35; 95 18.35 -1.2 12.35; 96 24.35 0 12.35; 97 24.35 4 12.35;
98 24.35 -1.2 12.35; 114 6.5 0 0; 115 12.65 0 0; 116 18.35 0 0; 117 24.35 0 0;
118 28.95 4 0; 119 28.95 -1.2 0; 120 28.95 4 7.19; 121 28.95 -1.2 7.19;
122 28.95 0 12.35; 123 28.95 4 12.35; 124 28.95 -1.2 12.35; 125 28.95 0 0;
126 34.95 4 0; 127 34.95 -1.2 0; 128 34.95 4 7.19; 129 34.95 -1.2 7.19;
130 34.95 0 12.35; 131 34.95 4 12.35; 132 34.95 -1.2 12.35; 133 34.95 0 0;
134 40.75 4 0; 135 40.75 -1.2 0; 136 40.75 4 7.19; 137 40.75 -1.2 7.19;
138 40.75 0 12.35; 139 40.75 4 12.35; 140 40.75 -1.2 12.35; 141 40.75 0 0;
142 46.01 4 0; 143 46.01 -1.2 0; 144 46.01 4 7.19; 145 46.01 -1.2 7.19;
146 46.01 0 12.35; 147 46.01 4 12.35; 148 46.01 -1.2 12.35; 149 46.01 0 0;
150 46.53 4 0; 151 46.53 0 0; 152 46.53 4 7.19; 153 46.53 -1.2 7.19;
154 46.53 0 12.35; 155 46.53 4 12.35; 156 46.53 -1.2 12.35; 157 46.53 -1.2 0;
158 51.79 4 0; 159 51.79 0 0; 160 51.79 4 7.19; 161 51.79 -1.2 7.19;
162 51.79 0 12.35; 163 51.79 4 12.35; 164 51.79 -1.2 12.35; 165 51.79 -1.2 0;
166 57.54 4 0; 167 57.54 0 0; 168 57.54 4 7.19; 169 57.54 -1.2 7.19;
170 57.54 0 12.35; 171 57.54 4 12.35; 172 57.54 -1.2 12.35; 173 57.54 -1.2 0;
174 63.29 4 0; 175 63.29 0 0; 176 63.29 4 7.19; 177 63.29 -1.2 7.19;
178 63.29 0 12.35; 179 63.29 4 12.35; 180 63.29 -1.2 12.35; 181 63.29 -1.2 0;
182 69.04 4 0; 183 69.04 0 0; 184 69.04 4 7.19; 185 69.04 -1.2 7.19;
186 69.04 0 12.35; 187 69.04 4 12.35; 188 69.04 -1.2 12.35; 189 69.04 -1.2 0;
190 74.84 4 0; 191 74.84 0 0; 192 74.84 4 7.19; 193 74.84 -1.2 7.19;
194 74.84 0 12.35; 195 74.84 4 12.35; 196 74.84 -1.2 12.35; 197 74.84 -1.2 0;
198 80.64 4 0; 199 80.64 0 0; 200 80.64 4 7.19; 201 80.64 -1.2 7.19;
202 80.64 0 12.35; 203 80.64 4 12.35; 204 80.64 -1.2 12.35; 205 80.64 -1.2 0;
206 85.84 4 0; 207 85.84 0 0; 208 85.84 4 7.19; 209 85.84 -1.2 7.19;
210 85.84 0 12.35; 211 85.84 4 12.35; 212 85.84 -1.2 12.35; 213 85.84 -1.2 0;
214 90.7 4 0; 215 90.7 0 0; 216 90.7 4 7.19; 217 90.7 -1.2 7.19;
218 90.7 0 12.35; 219 90.7 4 12.35; 220 90.7 -1.2 12.35; 221 90.7 -1.2 0;
222 94.6 4 0; 223 94.6 0 0; 224 94.6 4 7.19; 225 94.6 -1.2 7.19;
226 94.6 0 12.35; 227 94.6 4 12.35; 228 94.6 -1.2 12.35; 229 94.6 -1.2 0;
230 94.6 -1.01454e-009 7.19; 231 46.01 -1.01454e-009 7.19;
232 46.53 -1.01454e-009 7.19; 233 97.55 4 4.525; 234 97.55 -1.01454e-009 4.525;
235 97.55 -1.2 4.525; 236 97.55 4 9.625; 237 97.55 -1.01454e-009 9.625;
238 97.55 -1.2 9.625; 239 94.6 -7.20126e-010 4.525; 240 94.6 4 4.525;
241 94.6 4 9.625; 242 94.6 -5.35779e-010 9.625;
MEMBER INCIDENCES
70 42 43; 71 42 44; 73 43 46; 74 46 114; 77 46 49; 78 49 115; 81 49 52;
82 52 116; 85 52 55; 86 55 117; 96 42 63; 97 43 64; 110 63 64; 111 63 65;
113 64 67; 114 67 68; 117 67 70; 118 70 71; 121 70 73; 122 73 74; 125 73 76;
126 76 77; 136 63 84; 137 64 85; 150 84 85; 151 84 86; 152 84 87; 153 85 88;
154 87 88; 155 87 89; 156 87 90; 157 88 91; 158 90 91; 159 90 92; 160 90 93;
161 91 94; 162 93 94; 163 93 95; 164 93 96; 165 94 97; 166 96 97; 167 96 98;
190 114 47; 192 115 50; 194 116 53; 196 117 56; 201 42 114; 202 114 115;
203 115 116; 204 116 117; 205 55 118; 207 76 120; 209 96 122; 210 97 123;
212 117 125; 213 118 125; 214 120 121; 215 122 123; 216 122 124; 217 125 119;

218 118 126; 220 120 128; 222 122 130; 223 123 131; 225 125 133; 226 126 133;
227 128 129; 228 130 131; 229 130 132; 230 133 127; 231 126 134; 233 128 136;
235 130 138; 236 131 139; 238 133 141; 239 134 141; 240 136 137; 241 138 139;
242 138 140; 243 141 135; 244 134 142; 246 136 144; 248 138 146; 249 139 147;
251 141 149; 252 142 149; 253 144 231; 254 146 147; 255 146 148; 256 149 143;
257 150 151; 258 152 232; 259 154 155; 260 154 156; 261 151 157; 262 150 158;
263 151 159; 264 152 160; 266 154 162; 267 155 163; 270 158 159; 271 160 161;
272 162 163; 273 162 164; 274 159 165; 275 158 166; 276 159 167; 277 160 168;
279 162 170; 280 163 171; 283 166 167; 284 168 169; 285 170 171; 286 170 172;
287 167 173; 288 166 174; 289 167 175; 290 168 176; 292 170 178; 293 171 179;
296 174 175; 297 176 177; 298 178 179; 299 178 180; 300 175 181; 301 174 182;
302 175 183; 303 176 184; 305 178 186; 306 179 187; 309 182 183; 310 184 185;
311 186 187; 312 186 188; 313 183 189; 314 182 190; 315 183 191; 316 184 192;
318 186 194; 319 187 195; 322 190 191; 323 192 193; 324 194 195; 325 194 196;
326 191 197; 327 190 198; 328 191 199; 329 192 200; 330 194 202; 331 195 203;
332 198 199; 333 200 201; 334 202 203; 335 202 204; 336 199 205; 337 198 206;
338 199 207; 339 200 208; 340 202 210; 341 203 211; 342 206 207; 343 208 209;
344 210 211; 345 210 212; 346 207 213; 347 206 214; 348 207 215; 349 208 216;
350 210 218; 351 211 219; 352 214 215; 353 216 217; 354 218 219; 355 218 220;
356 215 221; 357 214 222; 358 215 223; 359 216 224; 360 218 226; 361 219 227;
362 222 223; 363 224 230; 364 226 227; 365 226 228; 366 223 229; 367 222 240;
368 224 241; 369 230 225; 370 223 239; 371 230 242; 372 142 144; 373 144 147;
374 150 152; 375 152 155; 376 231 145; 377 149 231; 378 232 153; 379 151 232;
380 231 146; 381 232 154; 382 233 234; 383 234 235; 384 236 237; 385 237 238;
386 239 230; 387 234 239; 388 240 224; 389 233 240; 390 241 227; 391 236 241;
392 233 236; 393 237 234; 394 242 226; 395 237 242; 396 240 239; 397 241 242;
398 142 150; 399 149 151; 400 146 154; 401 147 155;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 28000
G 9.28139e+006
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
70 74 78 82 86 110 150 154 158 162 166 213 215 226 228 239 241 252 254 257 -
259 270 272 283 285 296 298 309 311 322 324 332 334 342 344 352 354 -
362 TO 364 369 PRIS YD 0.55 ZD 0.3
MEMBER PROPERTY AMERICAN
113 117 121 125 207 220 233 246 264 277 290 303 316 329 339 349 -
359 PRIS YD 0.55 ZD 0.25

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

MEMBER PROPERTY AMERICAN
73 77 81 85 97 137 153 157 161 165 205 210 218 223 231 236 244 249 262 267 -
275 280 288 293 301 306 314 319 327 331 337 341 347 351 357 361 372 TO 375 -
392 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
114 118 122 126 214 227 240 253 258 271 284 297 310 323 333 343 -
353 PRIS YD 0.55 ZD 0.25
MEMBER PROPERTY AMERICAN
396 TO 398 401 PRIS YD 0.1 ZD 0.1
96 136 152 156 160 164 201 TO 204 209 212 222 225 235 238 248 251 263 266 -
276 279 289 292 302 305 315 318 328 330 338 340 348 350 358 360 367 368 370 -
371 377 379 TO 381 386 TO 388 390 393 TO 395 399 400 PRIS YD 0.6 ZD 0.25
389 391 PRIS YD 0.35 ZD 0.25
MEMBER PROPERTY AMERICAN
71 111 151 155 159 163 167 190 192 194 196 216 217 229 230 242 243 255 256 -
260 261 273 274 286 287 299 300 312 313 325 326 335 336 345 346 355 356 365 -
366 376 378 PRIS YD 0.55 ZD 0.35
MEMBER PROPERTY AMERICAN
382 TO 385 PRIS YD 0.4 ZD 0.25
CONSTANTS
BETA 90 MEMB 110 111 114 150 151 382 TO 385
MATERIAL CONCRETE ALL
SUPPORTS
44 47 50 53 56 65 68 71 74 77 86 89 92 95 98 119 121 124 127 129 132 135 137 -
140 143 145 148 153 156 157 161 164 165 169 172 173 177 180 181 185 188 189 -
193 196 197 201 204 205 209 212 213 217 220 221 225 228 229 235 238 FIXED
****
MEMBER CRACKED
73 77 81 85 96 97 113 117 121 125 136 137 152 153 156 157 160 161 164 165 -
201 TO 205 207 209 210 212 218 220 222 223 225 231 233 235 236 238 244 246 -
248 249 251 262 TO 264 266 267 275 TO 277 279 280 288 TO 290 292 293 301 -
302 TO 303 305 306 314 TO 316 318 319 327 TO 331 337 TO 341 347 TO 351 357 -
358 TO 361 367 368 370 TO 375 377 379 TO 381 386 TO 395 398 TO 400 -
401 REDUCTION RIX 0.350000 RIY 0.350000 RIZ 0.350000
70 71 74 78 82 86 110 111 114 118 122 126 150 151 154 155 158 159 162 163 -
166 167 190 192 194 196 213 TO 217 226 TO 230 239 TO 243 252 TO 261 -
270 TO 274 283 TO 287 296 TO 300 309 TO 313 322 TO 326 332 TO 336 -
342 TO 346 352 TO 356 362 TO 366 369 376 378 382 TO 385 396 -
397 REDUCTION RIX 0.700000 RIY 0.700000 RIZ 0.700000
MEMBER OFFSET
73 77 81 85 113 117 121 125 153 157 161 165 205 207 210 218 220 223 231 233 -
236 244 246 249 262 264 267 275 277 280 288 290 293 301 303 306 314 316 319 -
327 329 331 337 339 341 347 349 351 357 359 361 398 401 START 0.275 0 0
73 77 81 85 113 117 121 125 153 157 161 165 205 207 210 218 220 223 231 233 -
236 244 246 249 262 264 267 275 277 280 288 290 293 301 303 306 314 316 319 -

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

327 329 331 337 339 341 347 349 351 357 359 361 398 401 END -0.275 0 0
DEFINE IBC 2003
SDS 0.106 SD1 0.032 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 4
SELFWEIGHT 1
MEMBER WEIGHT
96 136 152 156 160 164 201 TO 204 209 212 222 225 235 238 248 251 263 266 -
276 279 289 292 302 305 315 318 328 330 338 340 348 350 358 360 370 371 386 -
387 393 TO 395 UNI 14
113 117 121 125 207 220 233 246 264 277 290 303 316 329 339 349 359 UNI 38.48
153 157 161 165 210 223 236 249 267 280 293 306 319 331 341 351 361 389 -
391 UNI 15
73 77 81 85 205 218 231 244 262 275 288 301 314 327 337 347 357 UNI 23.48
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 35.000 M 25.000 M 2.000 0.010 0 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.59699 1.59699 HEIG 0 4.572
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.500 M 100.000 M 15.000 M 2.000 0.010 1 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.200 0.550
!> END GENERATED DATA BLOCK
INT -0.934826 -0.934826 HEIG 0 4.5
TYPE 3 SIDEWARD X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.500 M 100.000 M 15.000 M 2.000 0.010 2 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.885 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.51878 -1.51878 HEIG 0 4.5
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.500 M 15.000 M 100.000 M 2.000 0.010 0 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.828 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.57461 1.57461 HEIG 0 4.572

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

```
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.500 M 15.000 M 100.000 M 2.000 0.010 1 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.828 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.25192 -1.25192 HEIG  0  4.5
TYPE 6 SIDEWARD Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.500 M 15.000 M 100.000 M 2.000 0.010 2 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.828 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.46705 -1.46705 HEIG  0  4.5
LOAD 1 LOADTYPE Seismic  TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic  TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
97 137 UMOM GZ 1
367 390 392 UMOM GZ -1
153 157 161 165 210 223 236 249 267 280 293 306 319 331 341 351 361 UMOM GX 1
73 77 81 85 205 218 231 244 262 275 288 301 314 327 337 347 357 UMOM GX -1
96 136 152 156 160 164 201 TO 204 209 212 222 225 235 238 248 251 263 266 -
276 279 289 292 302 305 315 318 328 330 338 340 348 350 358 360 370 371 386 -
387 393 TO 395 UNI GY -14
113 117 121 125 207 220 233 246 264 277 290 303 316 329 339 349 359 UNI GY -38
153 157 161 165 210 223 236 249 267 280 293 306 319 331 341 351 361 389 -
391 UNI GY -15
73 77 81 85 205 218 231 244 262 275 288 301 314 327 337 347 357 UNI GY -23
LOAD 4 LOADTYPE Live  TITLE LL
MEMBER LOAD
97 137 UMOM GZ 0.18
153 157 161 165 210 223 236 249 267 280 293 306 319 331 341 351 -
361 UMOM GX 0.18
367 390 392 UMOM GZ -0.18
73 77 81 85 205 218 231 244 262 275 288 301 314 327 337 347 357 UMOM GX -0.18
```


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

113 117 121 125 207 220 233 246 264 277 290 303 316 329 339 349 359 UNI GY -3
153 157 161 165 210 223 236 249 267 280 293 306 319 331 341 351 361 389 -
391 UNI GY -1.3
73 77 81 85 205 218 231 244 262 275 288 301 314 327 337 347 357 UNI GY -1.8
LOAD 5 LOADTYPE Wind TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 4
WIND LOAD -X 1 TYPE 2 YR 0 4
WIND LOAD -Z 1 TYPE 3 YR 0 4
WIND LOAD -Z -1 TYPE 3 YR 0 4
LOAD 6 LOADTYPE Wind TITLE WL+Z
WIND LOAD Z 1 TYPE 4 YR 0 4
WIND LOAD -Z 1 TYPE 5 YR 0 4
WIND LOAD -X 1 TYPE 6 YR 0 4
WIND LOAD -X -1 TYPE 6 YR 0 4
LOAD 7 LOADTYPE Wind TITLE WL-X
WIND LOAD X -1 TYPE 1 YR 0 4
WIND LOAD -X -1 TYPE 2 YR 0 4
WIND LOAD -Z 1 TYPE 3 YR 0 4
WIND LOAD -Z -1 TYPE 3 YR 0 4
LOAD 8 LOADTYPE Wind TITLE WL-Z
WIND LOAD Z -1 TYPE 4 YR 0 4
WIND LOAD -Z -1 TYPE 5 YR 0 4
WIND LOAD -X 1 TYPE 6 YR 0 4
WIND LOAD -X -1 TYPE 6 YR 0 4
LOAD 9 LOADTYPE Temperature TITLE TEMP
TEMPERATURE LOAD
70 73 74 77 78 81 82 85 86 97 110 113 117 121 125 137 150 153 154 157 158 -
161 162 165 166 205 207 210 213 215 218 220 223 226 228 231 233 236 239 241 -
244 246 249 252 TO 254 257 TO 259 262 264 267 270 272 275 277 280 283 285 -
288 290 293 296 298 301 303 306 309 311 314 316 319 322 324 327 329 331 332 -
334 337 339 341 342 344 347 349 351 352 354 357 359 361 TO 364 367 368 372 -
373 TO 375 382 384 388 TO 392 396 TO 398 401 TEMP 25 25 25
LOAD COMB 10 SLS: (DL)
3 1.0
LOAD COMB 11 SLS: (DL+T)
3 1.0
LOAD COMB 12 SLS: (D+LR)
3 1.0 4 1.0
LOAD COMB 13 SLS: (D+0.75LR)
3 1.0 4 0.75
LOAD COMB 14 SLS: (D+0.7EZ)
3 1.0 1 0.7
LOAD COMB 15 SLS: (D+0.7EX)
3 1.0 2 0.7
LOAD COMB 16 SLS: (D+0.75WX1+0.75LR)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

3 1.0 5 0.75 4 0.75
LOAD COMB 17 SLS: (D+0.75WZ1+0.75LR)
3 1.0 6 0.75 4 0.75
LOAD COMB 18 SLS: (0.6D+WX1)
3 0.6 5 1.0
LOAD COMB 19 SLS: (0.6D+WZ1)
3 0.6 6 1.0
LOAD COMB 20 SLS: (0.6D+0.7EX)
3 0.6 2 0.7
LOAD COMB 21 SLS: (0.6D+0.7EZ)
3 0.6 1 0.7
LOAD COMB 22 SLS: (D+WX1)
3 1.0 5 1.0
LOAD COMB 23 SLS: (D+WZ1)
3 1.0 6 1.0
LOAD COMB 24 ULS: (1.4 DL)
3 1.4
LOAD COMB 25 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5
LOAD COMB 26 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5
LOAD COMB 27 ULS: (1.2D+0.80WX1+1.6LR)
3 1.2 5 0.8 4 1.6
LOAD COMB 28 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 6 1.6 4 0.5
LOAD COMB 29 ULS: (1.2D+1.6WX1+0.5LR)
3 1.2 4 0.5 5 1.6
LOAD COMB 30 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 4 0.5 6 1.6
LOAD COMB 31 ULS: (1.2D+EX)
3 1.2 2 1.0
LOAD COMB 32 ULS: (1.2D+EZ)
3 1.2 1 1.0
LOAD COMB 33 ULS: (0.90D+1.6WX1)
3 0.9 5 1.6
LOAD COMB 34 ULS: (0.90D+1.6WZ1)
3 0.9 6 1.6
LOAD COMB 35 ULS: (0.90D+EZ)
3 0.9 1 1.0
LOAD COMB 36 ULS: (0.90D+EX)
3 0.9 2 1.0
LOAD COMB 39 SLS: (D+0.75WX2+0.75LR)
3 1.0 7 0.75 4 0.75
LOAD COMB 40 SLS: (D+0.75WZ2+0.75LR)
3 1.0 8 0.75 4 0.75

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

LOAD COMB 41 SLS: (0.6D+WX2)
3 0.6 7 1.0
LOAD COMB 42 SLS: (0.6D+WZ2)
3 0.6 8 1.0
LOAD COMB 45 SLS: (D+WX2)
3 1.0 7 1.0
LOAD COMB 46 SLS: (D+WZ2)
3 1.0 8 1.0
LOAD COMB 47 ULS: (1.2D+0.80WX2+1.6LR)
3 1.2 7 0.8 4 1.6
LOAD COMB 48 ULS: (1.2D+1.6WZ2+0.5LR)
3 1.2 8 1.6 4 0.5
LOAD COMB 49 ULS: (1.2D+1.6WX2+0.5LR)
3 1.2 4 0.5 7 1.6
LOAD COMB 50 ULS: (1.2D+1.6WZ2+0.5LR)
3 1.2 4 0.5 8 1.6
LOAD COMB 53 ULS: (0.90D+1.6WX2)
3 0.9 7 1.6
LOAD COMB 54 ULS: (0.90D+1.6WZ2)
3 0.9 8 1.6
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
FC 35000 ALL
CLB 0.04 ALL
CLS 0.04 ALL
CLT 0.04 ALL
MINSEC 10 ALL
MINMAIN 20 MEMB 70 71 73 74 77 78 81 82 86 110 111 113 114 117 118 121 122 -
125 126 150 151 153 TO 155 157 TO 159 161 TO 163 165 TO 167 190 192 194 196 -
213 TO 217 226 TO 230 239 TO 243 252 TO 261 270 TO 274 283 TO 287 -
296 TO 300 309 TO 313 322 TO 326 332 TO 336 342 TO 346 352 TO 356 -
362 TO 366 369 376 378 382 TO 385
SFACE 0.5 MEMB 73 77 81 85 153 157 161 165 205 210 218 223 231 236 244 249 -
262 267 275 280 288 293 301 306 314 319 327 331 337 341 347 351 357 361 398 -
401
EFACE 0.5 MEMB 73 77 81 85 153 157 161 165 205 210 218 223 231 236 244 249 -
262 267 275 280 288 293 301 306 314 319 327 331 337 341 347 351 357 361 398 -
401
SFACE 0.55 MEMB 113 117 121 125 207 220 233 246 264 277 290 303 316 329 339 -
349 359
EFACE 0.55 MEMB 113 117 121 125 207 220 233 246 264 277 290 303 316 329 339 -
349 359
DESIGN COLUMN 70 71 74 78 82 86 110 111 114 118 122 126 150 151 154 155 158 -
159 162 163 166 167 190 192 194 196 213 TO 217 226 TO 230 239 TO 243 252 -

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

253 TO 261 270 TO 274 283 TO 287 296 TO 300 309 TO 313 322 TO 326 332 TO 336 -
342 TO 346 352 TO 356 362 TO 366 369 376 378 382 TO 385
DESIGN BEAM 73 77 81 85 96 97 113 117 121 125 136 137 152 153 156 157 160 -
161 164 165 201 TO 205 207 209 210 212 218 220 222 223 225 231 233 235 236 -
238 244 246 248 249 251 262 TO 264 266 267 275 TO 277 279 280 288 TO 290 -
292 293 301 TO 303 305 306 314 TO 316 318 319 327 TO 331 337 TO 341 -
347 TO 351 357 TO 361 367 368 370 TO 375 377 379 TO 381 386 TO 395 -
398 TO 401
END CONCRETE DESIGN
FINISH

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

ATTACHMENT 2 FOUNDATION DESIGN

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

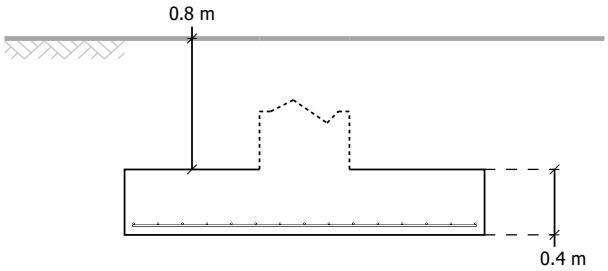
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Isolated Footing Design(ACI 318-08)

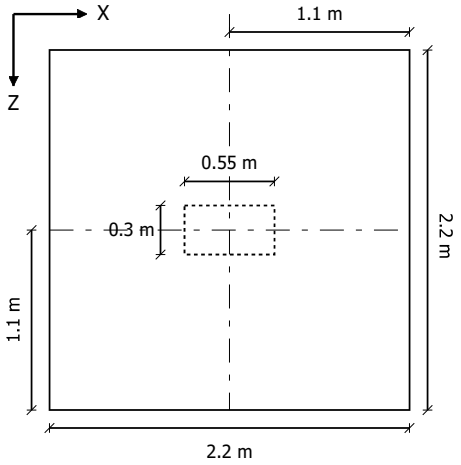
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
50	1	2.200m	2.200m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
50	15 - ϕ12	15 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F1



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 2200.000mm
Footing Width - Z (Fw) : 2200.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.550m

Column Width - Z (B_{col}) : 0.300m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 100.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 2.200m

Initial Width (W_o) = 2.200m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	263.006	2.946	0.007	-4.979	-1.752
11	263.006	2.946	0.007	-4.979	-1.752
12	272.708	2.653	0.005	-5.899	-1.642
13	270.283	2.726	0.005	-5.669	-1.670
14	263.002	3.043	5.560	18.279	-1.951
15	262.768	6.588	0.007	-5.004	-8.901
16	270.218	4.762	-27.153	-88.884	-4.855
17	270.335	0.922	28.146	80.816	1.109
18	157.717	4.483	-36.207	-113.940	-5.298
19	157.873	-0.637	37.525	112.326	2.653

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
20	157.566	5.409	0.004	-3.012	-8.200
21	157.800	1.865	5.557	20.271	-1.250
22	262.919	5.661	-36.204	-115.932	-5.999
23	263.076	0.541	37.528	110.334	1.952
39	270.319	1.431	-27.153	-88.868	0.393
40	270.332	0.849	-22.365	-74.457	1.254
41	157.852	0.041	-36.207	-113.919	1.700
42	157.869	-0.735	-29.822	-94.704	2.847
45	263.055	1.219	-36.204	-115.911	0.999
46	263.072	0.444	-29.820	-96.695	2.146

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	368.209	4.125	0.010	-6.970	-2.453
25	373.060	3.978	0.009	-7.430	-2.398
26	373.060	3.978	0.009	-7.430	-2.398
27	331.061	5.238	-28.965	-96.210	-5.324
28	320.570	-0.460	60.040	178.066	3.880
29	320.319	7.733	-57.931	-183.959	-8.842
30	320.570	-0.460	60.040	178.066	3.880
31	315.267	8.738	0.008	-6.010	-12.315
32	315.602	3.674	7.941	27.251	-2.387
33	236.566	6.996	-57.931	-182.006	-8.371
34	236.817	-1.197	60.039	180.020	4.350
35	236.700	2.790	7.939	28.745	-1.861
36	236.365	7.854	0.006	-4.517	-11.790
47	331.170	1.684	-28.964	-96.193	0.274
48	320.563	-0.615	-47.715	-153.182	4.189
49	320.536	0.625	-57.930	-183.926	2.354
50	320.563	-0.615	-47.715	-153.182	4.189
53	236.783	-0.112	-57.931	-181.972	2.824
54	236.810	-1.353	-47.716	-151.228	4.660

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 4.840m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 2.032m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 2.200 m Governing Load Case : # 10

Width (W_2) = 2.200 m Governing Load Case : # 10

Depth (D_2) = 0.400 m Governing Load Case : # 49

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 4.840 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 46.462 kN

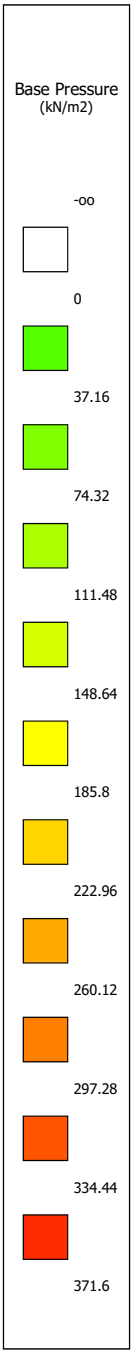
Gross Soil Bearing Capacity = 371.60 kN/m2

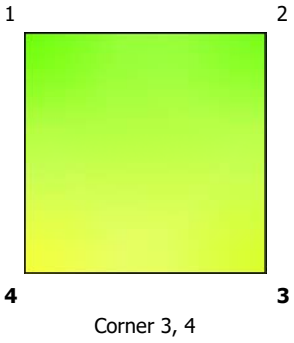
Soil Weight On Top Of Footing = 67.318 kN

Pressures at 4 Corners

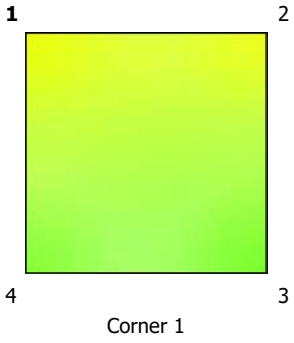
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

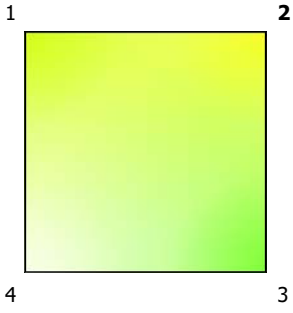




Corner 3, 4

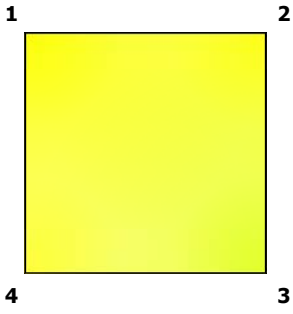


Corner 1



Corner 2

Governing Base Pressure Contour



Corner 1, 2, 3, 4

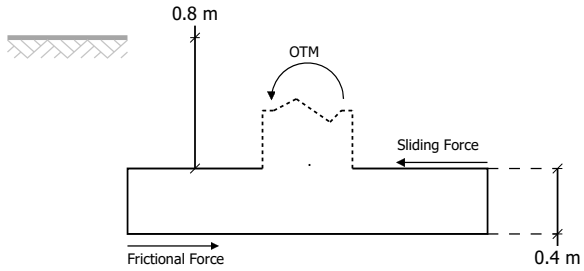
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
45	**151.6219**	151.0458	4.0949	4.6709	0.000
22	146.6614	**155.9738**	8.9993	-0.3130	0.000
23	8.2095	6.2530	**147.5161**	149.4726	0.000
23	8.2095	6.2530	147.5161	**149.4726**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
45	**151.6219**	151.0458	4.0949	4.6709
22	146.6616	**155.9739**	8.9991	0.0000
23	8.2095	6.2530	**147.5161**	149.4726
23	8.2095	6.2530	147.5161	**149.4726**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	63.945	26075.171	90.432	1.500	83.678	141.506	1.500
11	63.945	26075.171	90.432	1.500	83.678	141.506	1.500
12	72.846	42110.310	103.020	1.500	72.411	157.335	1.500
13	70.441	36591.357	99.619	1.500	74.884	153.133	1.500
14	61.909	33.883	42.034	1.500	20.307	130.878	1.500
15	28.579	28341.718	40.417	1.500	83.199	35.923	1.500
16	40.315	7.071	9.849	1.500	4.254	62.519	1.500
17	208.253	6.824	9.645	1.500	4.610	571.510	1.500
18	30.282	3.749	5.262	1.500	2.340	42.147	1.500
19	213.069	3.620	5.118	1.500	2.362	102.822	1.500
20	25.081	36150.493	35.471	1.500	99.762	28.820	1.500
21	72.827	24.435	32.761	1.500	13.365	149.759	1.500
22	33.269	5.202	7.269	1.500	3.192	50.173	1.500
23	348.301	5.021	7.100	1.500	3.322	238.903	1.500
39	134.230	7.073	9.989	1.500	4.256	2361.613	1.500
40	226.168	8.587	12.136	1.500	5.089	462.447	1.500
41	3350.338	3.751	5.305	1.500	2.342	177.627	1.500
42	184.823	4.554	6.439	1.500	2.820	95.206	1.500
45	154.566	5.204	7.356	1.500	3.194	811.341	1.500
46	424.785	6.319	8.935	1.500	3.834	210.675	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 20

Governing Disturbing Force : 5.409kN

Governing Restoring Force : 135.673kN

Minimum Sliding Ratio for the Critical Load Case : 25.081

Critical Load Case for Overturning about X-Direction : 18

Governing Overturning Moment : -128.423kNm

Governing Resisting Moment : 300.541kNm

Minimum Overturning Ratio for the Critical Load Case : 2.340

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 19

Governing Disturbing Force : 37.525kN

Governing Restoring Force : 135.827kN

Minimum Sliding Ratio for the Critical Load Case : 3.620

Critical Load Case for Overturning about Z-Direction : 20

Governing Overturning Moment : -10.364kNm

Governing Resisting Moment : 298.692kNm

Minimum Overturning Ratio for the Critical Load Case : 28.820

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 19

Governing Disturbing Force : 37.530kN

Governing Restoring Force : 192.088kN

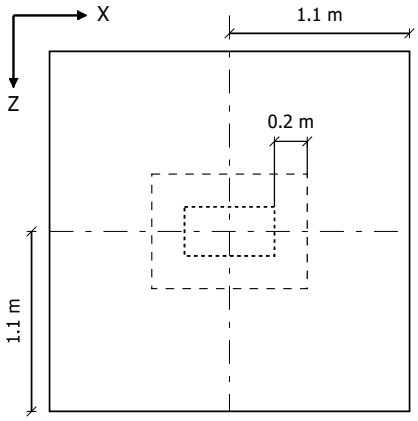
Minimum Sliding Ratio for the Critical Load Case : 5.118

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 1.833

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 451.742kN, Load Case # 29

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.076m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2173.720kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3364.859kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2079.210kN

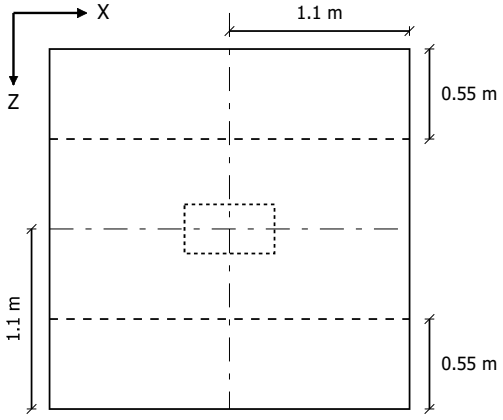
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1559.408kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 743.541kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.606m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

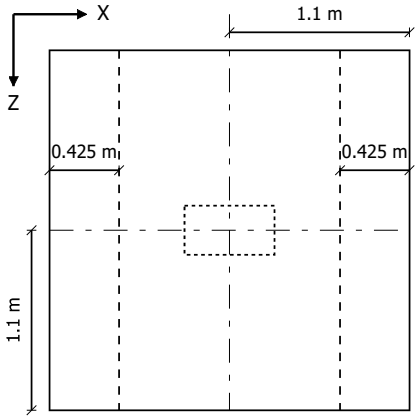
From above calculations, 0.75 X V_c = 557.655 kN

Critical load case for V_{ux} is # 49 $V_{ux} = V_{ux}|_{x=D_x}$ = 232.361 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 743.541 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.481 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

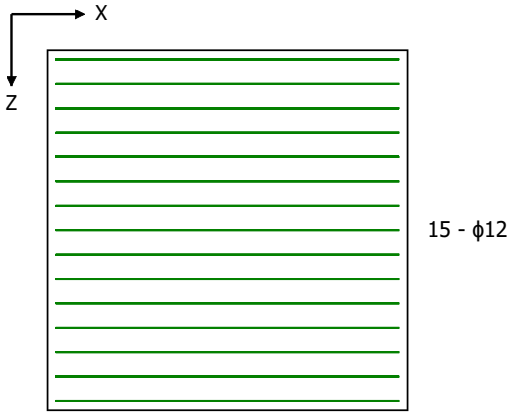
From above calculations, 0.75 X V_c = 557.655 kN

Critical load case for V_{uz} is # 25 $V_{uz} = V_{uz}|_{z=D_z} =$ 108.300 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 25

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00155
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.375 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	76.570 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	85.078 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00078
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00067
Since	$\rho = \rho_{min}$	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1362.928 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 1st

Max spacing for Cracking Consideration = 250.272mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover}$ = 0.775 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 15

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1696.448 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00224

From ACI Cl.7.6.1, minimum req'd clear distance between bars

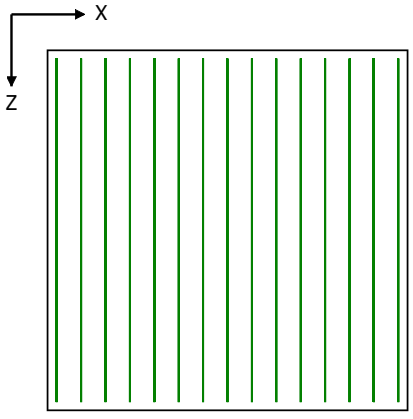
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.245

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



15 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 28

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl.7.12.2, ρ_{min} =		0.00162	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z =	$0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ =	1.250	m
Ultimate moment,	$M_u\big\vert_{x=D_x}$ =	176.885	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	196.539	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00196	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00162	
Since	$\rho_{min} < \rho < \rho_{max}$	OK	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1429.255	mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 3rd

Max spacing for Cracking Consideration = 250.272mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times \left(L - D_{col}\right) - C_{cover}$ =	0.900	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	15	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

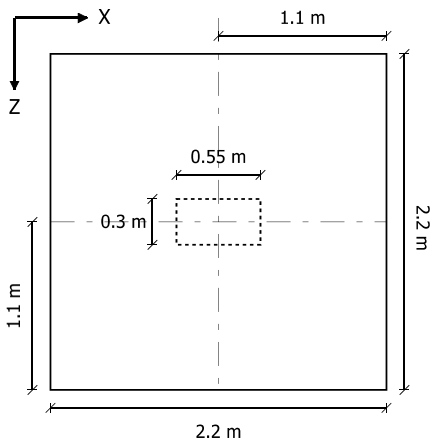
Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1696.448	mm2
d_{eff} =	D - C_{cover} - 1.5 X (dia. of one bar) =	0.332	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ =	0.00232	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

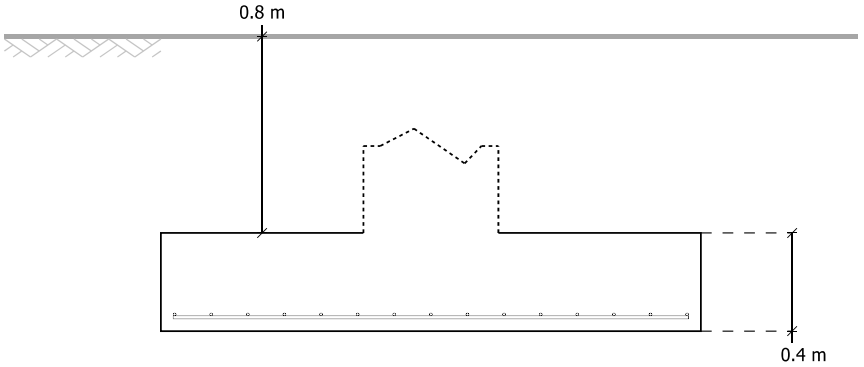
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.187

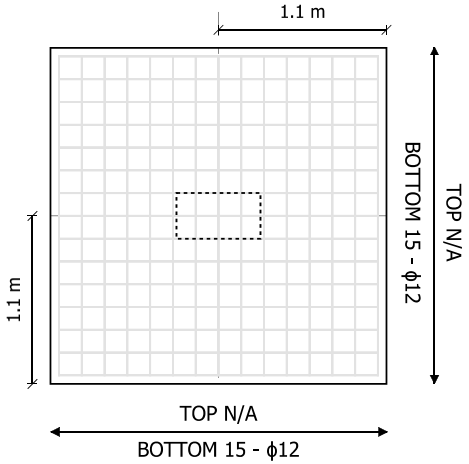
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	2.2 m	2.2 m	0.4 m	N/A	N/A	15 - ϕ12	15 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

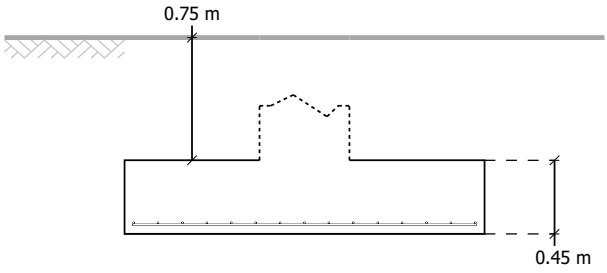
Print Calculation Sheet

Isolated Footing Design(ACI 318-08)

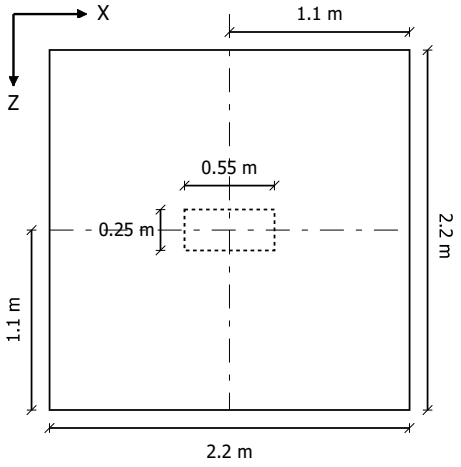
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
68	1	2.200m	2.200m	0.450m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
68	15 - ϕ12	15 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F2



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 450.000mm
Footing Length - X (Fl) : 2200.000mm
Footing Width - Z (Fw) : 2200.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.550m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 100.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 750.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 2.200m

Initial Width (W_o) = 2.200m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	278.682	8.073	-0.009	-0.061	-12.652
11	278.682	8.073	-0.009	-0.061	-12.652
12	297.421	8.614	-0.009	-0.062	-13.486
13	292.736	8.479	-0.009	-0.061	-13.277
14	278.670	8.141	6.341	29.498	-12.860
15	278.232	14.037	0.005	-0.005	-30.342
16	292.179	10.608	-0.019	-0.092	-19.773
17	293.227	6.629	0.073	0.493	-7.634
18	166.466	7.683	-0.019	-0.077	-16.253
19	167.863	2.377	0.104	0.703	-0.067

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
20	166.759	10.808	0.009	0.020	-25.282
21	167.197	4.912	6.345	29.522	-7.799
22	277.938	10.912	-0.022	-0.102	-21.314
23	279.336	5.606	0.100	0.679	-5.127
39	293.037	7.382	-0.034	-0.149	-9.931
40	293.245	6.576	-0.104	-0.670	-7.469
41	167.610	3.382	-0.038	-0.153	-3.130
42	167.887	2.307	-0.132	-0.848	0.153
45	279.083	6.611	-0.042	-0.177	-8.191
46	279.359	5.536	-0.136	-0.872	-4.908

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	390.155	11.302	-0.013	-0.086	-17.713
25	399.524	11.573	-0.013	-0.086	-18.130
26	399.524	11.573	-0.013	-0.086	-18.130
27	363.806	12.825	-0.022	-0.106	-23.445
28	344.835	6.011	0.164	1.110	-3.560
29	342.598	14.501	-0.032	-0.138	-29.458
30	344.835	6.011	0.164	1.110	-3.560
31	333.776	18.208	0.010	0.007	-40.454
32	334.401	9.785	9.061	42.154	-15.479
33	249.624	11.808	-0.029	-0.120	-25.245
34	251.860	3.318	0.167	1.129	0.653
35	250.797	7.363	9.064	42.172	-11.684
36	250.171	15.786	0.012	0.026	-36.659
47	364.722	9.384	-0.037	-0.167	-12.947
48	344.872	5.899	-0.214	-1.371	-3.209
49	344.429	7.619	-0.063	-0.259	-8.461
50	344.872	5.899	-0.214	-1.371	-3.209
53	251.455	4.926	-0.060	-0.241	-4.248
54	251.898	3.207	-0.211	-1.353	1.004

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 4.840m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.420m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 2.200 m Governing Load Case : # 10

Width (W_2) = 2.200 m Governing Load Case : # 10

Depth (D_2) = 0.450 m Governing Load Case : # 32

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 4.840 m^2

Final Soil Height = 0.750 m

Footing Self Weight = 52.270 kN

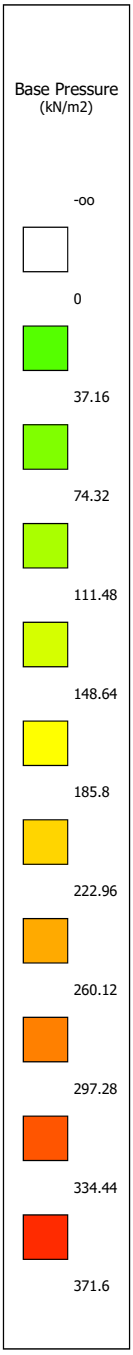
Gross Soil Bearing Capacity = 371.60 kN/m2

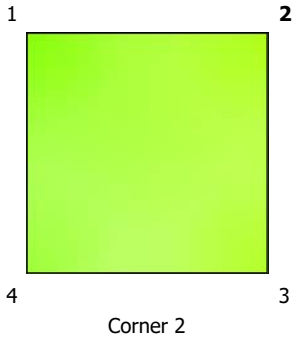
Soil Weight On Top Of Footing = 63.482 kN

Pressures at 4 Corners

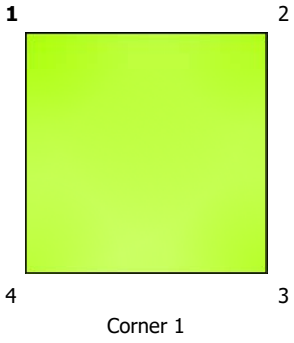
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

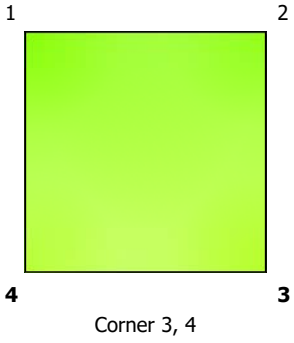




Corner 2

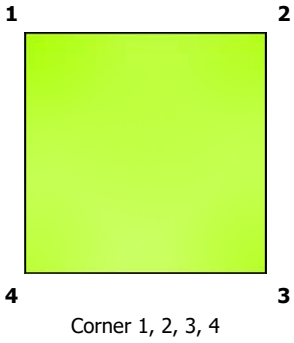


Corner 1



Corner 3, 4

Governing Base Pressure Contour



Corner 1, 2, 3, 4

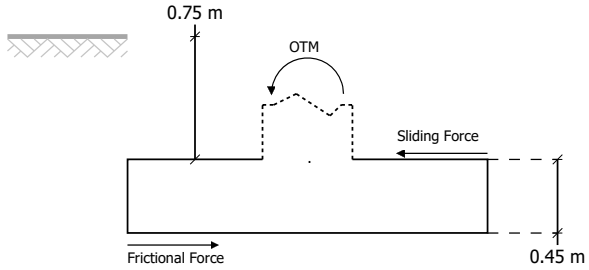
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
40	**79.0308**	90.7837	89.9759	78.2231	0.000
15	60.7457	**102.0600**	102.0575	60.7431	0.000
14	53.9513	72.5731	**109.0329**	90.4111	0.000
14	53.9513	72.5731	109.0329	**90.4111**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
40	**79.0308**	90.7837	89.9759	78.2231
15	60.7457	**102.0600**	102.0575	60.7431
14	53.9513	72.5731	**109.0329**	90.4111
14	53.9513	72.5731	109.0329	**90.4111**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	24.430	21676.937	34.549	1.500	6654.642	26.630	1.500
11	24.430	21676.937	34.549	1.500	6654.642	26.630	1.500
12	23.982	22239.901	33.915	1.500	6942.395	26.165	1.500
13	24.088	22101.939	34.066	1.500	6870.706	26.276	1.500
14	24.223	31.099	27.026	1.500	13.459	26.245	1.500
15	14.034	37737.898	19.847	1.500	190764.750	11.816	1.500
16	19.227	10606.615	27.191	1.500	4488.640	18.272	1.500
17	30.849	2805.875	43.625	1.500	857.874	42.355	1.500
18	18.367	7514.972	25.975	1.500	3648.633	15.740	1.500
19	59.669	1363.088	84.305	1.500	418.125	274.436	1.500
20	13.070	15944.852	18.483	1.500	13071.389	10.302	1.500
21	28.801	22.297	24.934	1.500	9.661	31.074	1.500
22	18.040	8781.451	25.512	1.500	3891.419	16.506	1.500
23	35.240	1967.666	49.829	1.500	602.674	56.783	1.500
39	27.687	6082.621	39.154	1.500	2757.076	33.913	1.500
40	31.096	1959.827	43.971	1.500	629.915	43.121	1.500
41	41.898	3734.239	59.248	1.500	1843.000	66.967	1.500
42	61.475	1072.235	86.797	1.500	345.586	352.192	1.500
45	29.863	4747.887	42.232	1.500	2222.196	38.880	1.500
46	35.685	1453.641	50.451	1.500	467.243	58.711	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 20

Governing Disturbing Force : 10.808kN

Governing Restoring Force : 141.255kN

Minimum Sliding Ratio for the Critical Load Case : 13.070

Critical Load Case for Overturning about X-Direction : 21

Governing Overturning Moment : 32.378kNm

Governing Resisting Moment : 312.816kNm

Minimum Overturning Ratio for the Critical Load Case : 9.661

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 21

Governing Disturbing Force : 6.345kN

Governing Restoring Force : 141.474kN

Minimum Sliding Ratio for the Critical Load Case : 22.297

Critical Load Case for Overturning about Z-Direction : 20

Governing Overturning Moment : -30.145kNm

Governing Resisting Moment : 310.552kNm

Minimum Overturning Ratio for the Critical Load Case : 10.302

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 20

Governing Disturbing Force : 10.808kN

Governing Restoring Force : 199.765kN

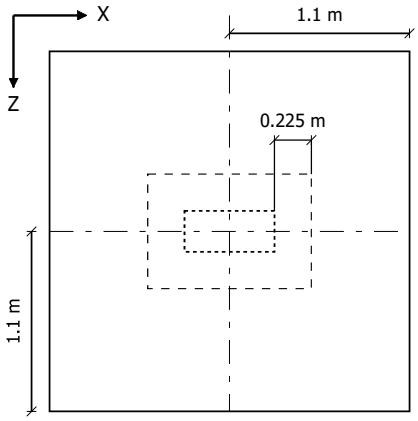
Minimum Sliding Ratio for the Critical Load Case : 18.483

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.450m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.394m

For rectangular column, β_c = B_{col} / D_{col} = 2.200

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 450.554kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.176m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2347.075kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 4279.745kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2458.840kN

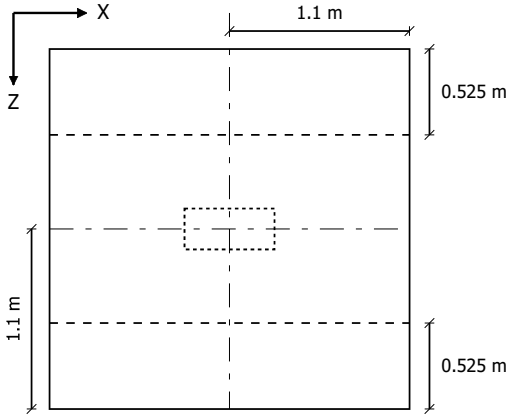
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1760.306kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 851.613kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.581m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

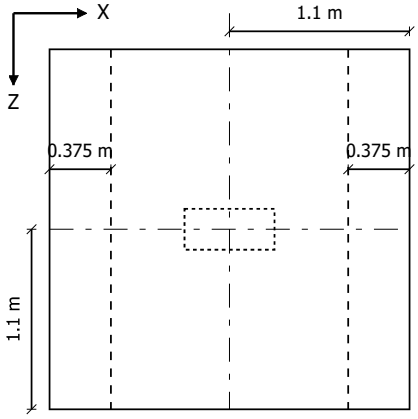
From above calculations, 0.75 X V_c = 638.710 kN

Critical load case for V_{ux} is # 32 $V_{ux} = V_{ux}|_{x=D_x}$ = 143.386 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 851.613 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.431 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

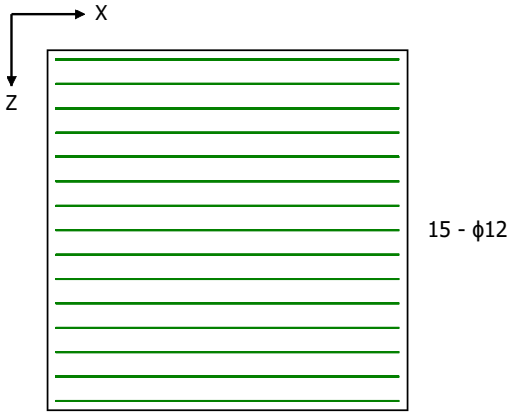
From above calculations, 0.75 X V_c = 638.710 kN

Critical load case for V_{uz} is # 25 $V_{uz} = V_{uz}|_{z=D_z} =$ 110.974 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 25

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.394 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00160
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.375 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	87.098 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	96.776 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00068
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00059
Since	$\rho = \rho_{min}$	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1580.784 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 1st

Max spacing for Cracking Consideration = 250.272mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover}$ = 0.775 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 15

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1696.448 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.394 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00196

From ACI Cl.7.6.1, minimum req'd clear distance between bars

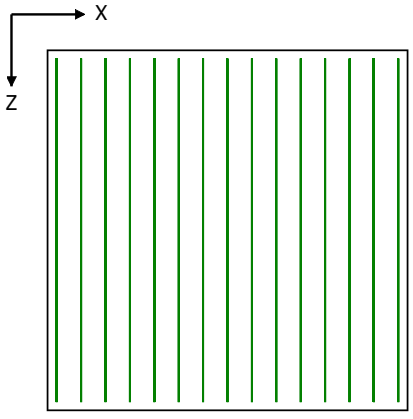
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.073

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



15 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 32

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.382 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00166

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.225 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 116.460 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 129.399 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00097

(Based on gross depth) ρ x d_{eff} / Depth = 0.00082

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1646.513 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 100.000mm

Selected spacing (S) = 149.143mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 3rd

Max spacing for Cracking Consideration = 250.272mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 145mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.925 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 15

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1696.448 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.382 m

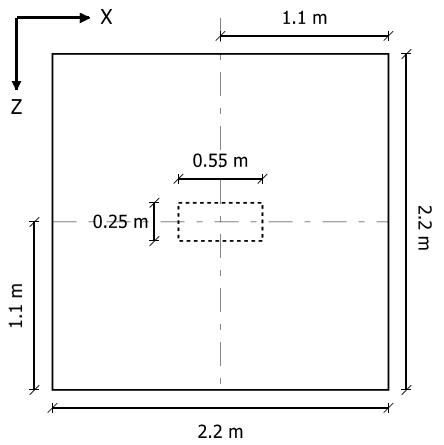
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00202

From ACI Cl.7.6.1, minimum req'd clear distance between bars

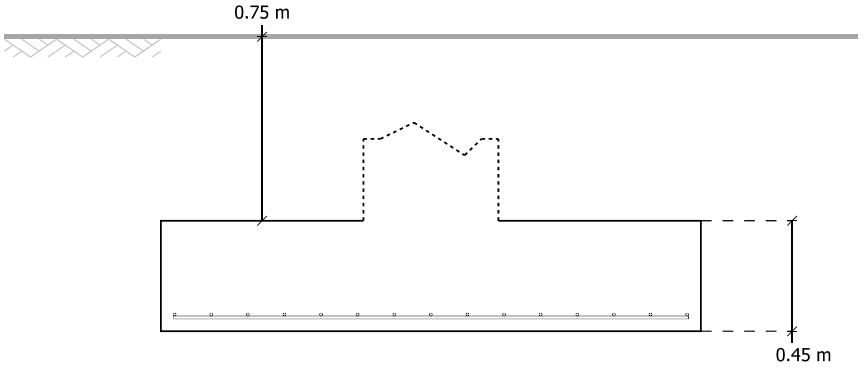
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.030

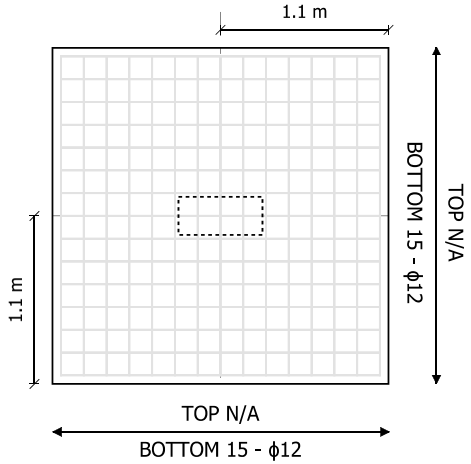
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
Foundation	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
	2.2 m	2.2 m	0.45 m	N/A	N/A	15 - ϕ12	15 - ϕ12
Pedestal	l	b	h	Main Reinforcement		Trans Reinforcement	
	N/A	N/A	N/A	N/A		N/A	

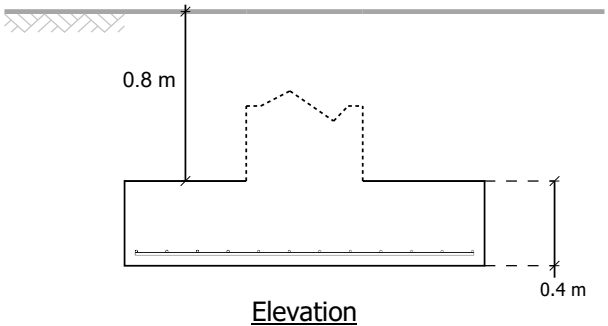
Print Calculation Sheet

Isolated Footing Design(ACI 318-08)

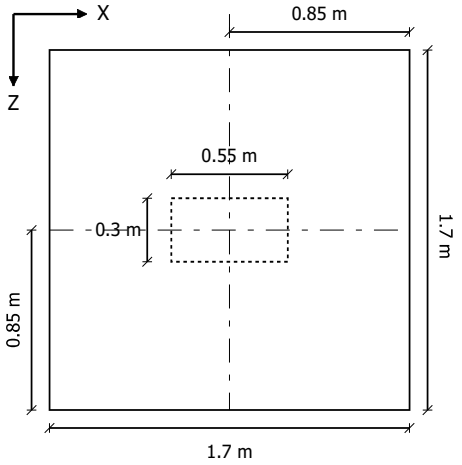
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
44	1	1.700m	1.700m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
44	12 - ϕ12	12 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F3



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1700.000mm
Footing Width - Z (Fw) : 1700.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.550m

Column Width - Z (B_{col}) : 0.300m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 100.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.700m

Initial Width (W_o) = 1.700m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	219.688	-21.931	-47.372	-13.427	7.877
11	219.688	-21.931	-47.372	-13.427	7.877
12	224.896	-21.552	-47.457	-13.428	8.569
13	223.594	-21.647	-47.436	-13.428	8.396
14	216.372	-21.820	-44.152	-7.211	7.632
15	218.163	-18.841	-47.375	-13.435	0.661
16	223.505	-16.971	-47.850	-14.070	2.518
17	219.927	-25.914	-41.369	-4.001	13.681
18	131.694	-6.925	-28.976	-8.912	-3.110
19	126.923	-18.848	-20.334	4.514	11.774

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
20	130.288	-10.069	-28.426	-8.064	-2.490
21	128.497	-13.047	-25.203	-1.840	4.482
22	219.569	-15.697	-47.925	-14.283	0.040
23	214.798	-27.621	-39.283	-0.857	14.924
39	224.535	-24.575	-47.846	-14.061	12.168
40	228.261	-26.036	-53.411	-22.710	13.918
41	133.067	-17.063	-28.971	-8.901	9.756
42	138.035	-19.011	-36.390	-20.433	12.089
45	220.942	-25.835	-47.920	-14.272	12.906
46	225.910	-27.783	-55.339	-25.804	15.239

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	307.563	-30.703	-66.321	-18.798	11.027
25	310.167	-30.514	-66.364	-18.799	11.373
26	310.167	-30.514	-66.364	-18.799	11.373
27	271.864	-20.724	-57.425	-16.799	4.290
28	258.405	-35.231	-43.946	3.999	21.074
29	266.040	-16.154	-57.773	-17.482	-2.740
30	258.405	-35.231	-43.946	3.999	21.074
31	261.447	-21.903	-56.851	-16.123	-0.856
32	258.889	-26.158	-52.247	-7.232	9.103
33	197.529	-9.764	-43.519	-13.453	-5.449
34	189.895	-28.842	-29.692	8.027	18.365
35	192.982	-19.579	-38.035	-3.204	6.740
36	195.541	-15.324	-42.639	-12.095	-3.219
47	272.962	-28.834	-57.420	-16.790	14.583
48	276.185	-35.492	-69.635	-35.915	21.578
49	268.236	-32.375	-57.765	-17.464	17.845
50	276.185	-35.492	-69.635	-35.915	21.578
53	199.725	-25.985	-43.511	-13.435	15.136
54	207.674	-29.102	-55.381	-31.887	18.869

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.890m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.494m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.700 m Governing Load Case : # 10

Width (W_2) = 1.700 m Governing Load Case : # 10

Depth (D_2) = 0.400 m Governing Load Case : # 50

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.890 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 27.743 kN

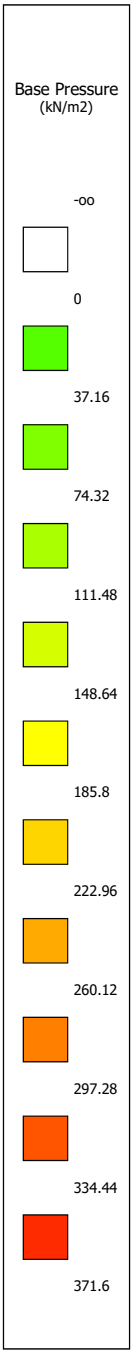
Gross Soil Bearing Capacity = 371.60 kN/m2

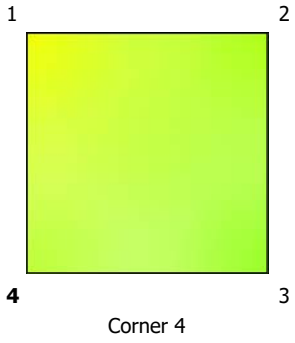
Soil Weight On Top Of Footing = 39.239 kN

Pressures at 4 Corners

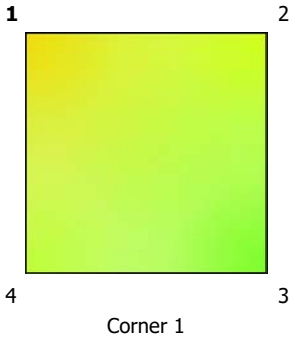
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

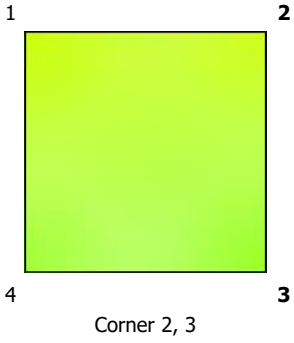




Corner 4

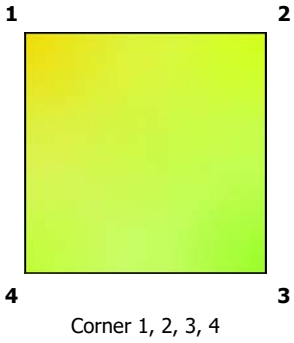


Corner 1



Corner 2, 3

Governing Base Pressure Contour



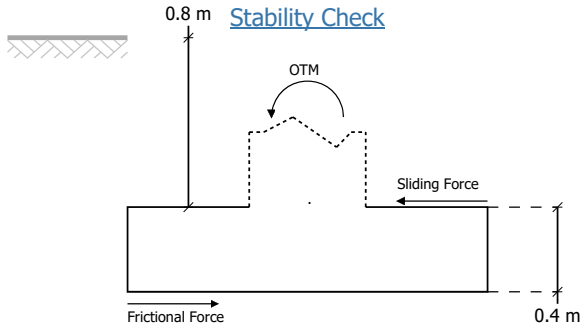
Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
46	**192.0765**	127.7091	10.6166	74.9840	0.000
22	147.7244	**132.2898**	50.5807	66.0152	0.000
22	147.7244	132.2898	**50.5807**	66.0152	0.000
23	149.4579	86.0188	45.5451	**108.9841**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
46	**192.0765**	127.7091	10.6166	74.9840
22	147.7244	**132.2898**	50.5807	66.0152
22	147.7244	132.2898	**50.5807**	66.0152
23	149.4579	86.0188	45.5451	**108.9841**



Stability Check

-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	6.536	3.026	3.883	1.500	7.571	14.646	1.500
11	6.536	3.026	3.883	1.500	7.571	14.646	1.500
12	6.772	3.075	3.960	1.500	7.700	14.443	1.500
13	6.712	3.063	3.941	1.500	7.668	14.492	1.500
14	6.493	3.209	4.068	1.500	9.743	14.732	1.500
15	7.567	3.009	3.955	1.500	7.529	29.587	1.500
16	8.558	3.035	4.046	1.500	7.479	26.548	1.500
17	5.536	3.468	4.156	1.500	11.940	10.148	1.500
18	14.345	3.428	4.716	1.500	8.308	496.434	1.500
19	5.144	4.768	4.945	1.500	45.935	8.543	1.500
20	9.796	3.470	4.625	1.500	8.704	109.140	1.500
21	7.491	3.878	4.870	1.500	14.061	17.146	1.500
22	9.127	2.990	4.018	1.500	7.325	38.571	1.500
23	5.101	3.587	4.149	1.500	14.543	9.228	1.500
39	5.931	3.046	3.832	1.500	7.508	11.272	1.500
40	5.670	2.764	3.514	1.500	5.727	10.321	1.500
41	5.862	3.453	4.207	1.500	8.371	10.265	1.500
42	5.392	2.817	3.531	1.500	5.023	8.857	1.500
45	5.572	3.004	3.740	1.500	7.363	10.538	1.500
46	5.271	2.646	3.345	1.500	5.224	9.453	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 23

Governing Disturbing Force : -27.621kN

Governing Restoring Force : 140.890kN

Minimum Sliding Ratio for the Critical Load Case : 5.101

Critical Load Case for Overturning about X-Direction : 42

Governing Overturning Moment : -34.988kNm

Governing Resisting Moment : 175.729kNm

Minimum Overturning Ratio for the Critical Load Case : 5.023

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 46

Governing Disturbing Force : -55.339kN

Governing Restoring Force : 146.446kN

Minimum Sliding Ratio for the Critical Load Case : 2.646

Critical Load Case for Overturning about Z-Direction : 19

Governing Overturning Moment : 19.313kNm

Governing Resisting Moment : 164.984kNm

Minimum Overturning Ratio for the Critical Load Case : 8.543

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 46

Governing Disturbing Force : 61.922kN

Governing Restoring Force : 207.106kN

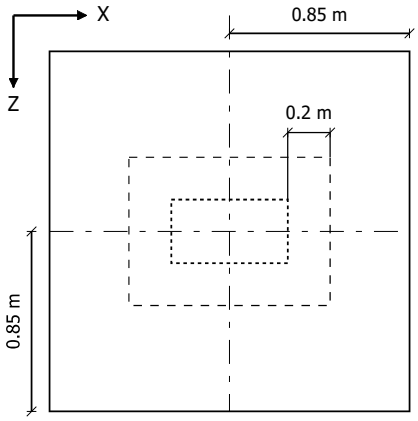
Minimum Sliding Ratio for the Critical Load Case : 3.345

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 1.833

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 302.015kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.076m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2173.720kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3364.859kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2079.210kN

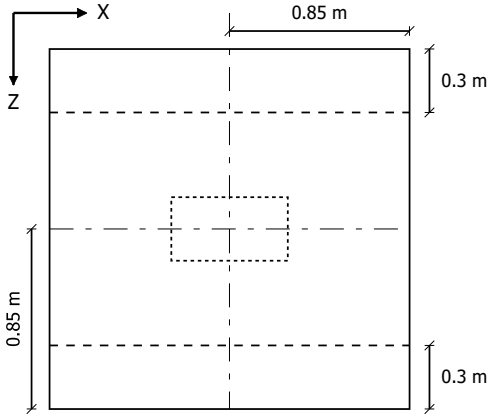
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1559.408kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 574.554kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.356m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

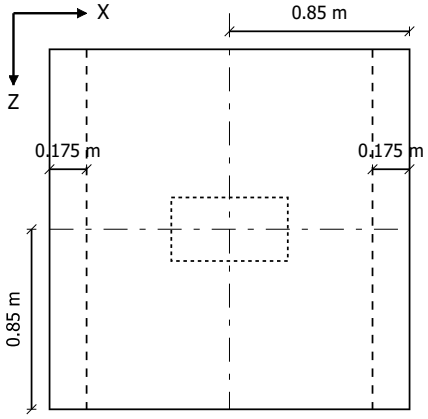
From above calculations, 0.75 X V_c = 430.916 kN

Critical load case for V_{ux} is # 48 $V_{ux} = V_{ux}|_{x=D_x}$ = 109.126 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 574.554 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.231 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

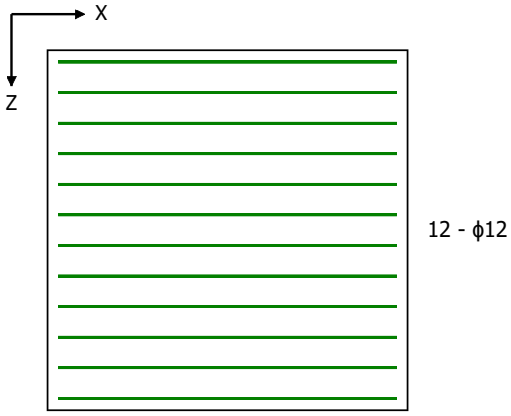
From above calculations, 0.75 X V_c = 430.916 kN

Critical load case for V_{uz} is # 50 $V_{uz} = V_{uz}|_{z=D_z} =$ 61.456 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 25

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00166
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.575 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	42.940 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	47.712 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00057
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00049
Since	$\rho = \rho_{min}$	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1131.328 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 1st

Max spacing for Cracking Consideration = 250.272mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot -C_{cover}$ = 0.525 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 12

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1357.159 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00232

From ACI Cl.7.6.1, minimum req'd clear distance between bars

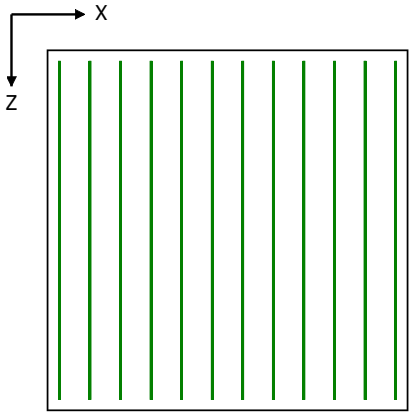
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.200

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



12 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 48

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00171

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.700 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 72.980 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 81.089 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00104

(Based on gross depth) ρ x d_{eff} / Depth = 0.00086

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1165.706 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 3rd

Max spacing for Cracking Consideration = 250.272mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.650 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 12

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1357.159 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

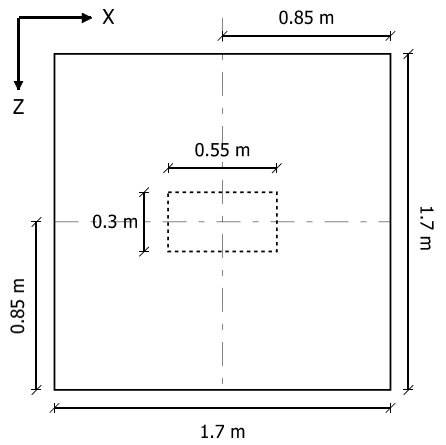
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00240

From ACI Cl.7.6.1, minimum req'd clear distance between bars

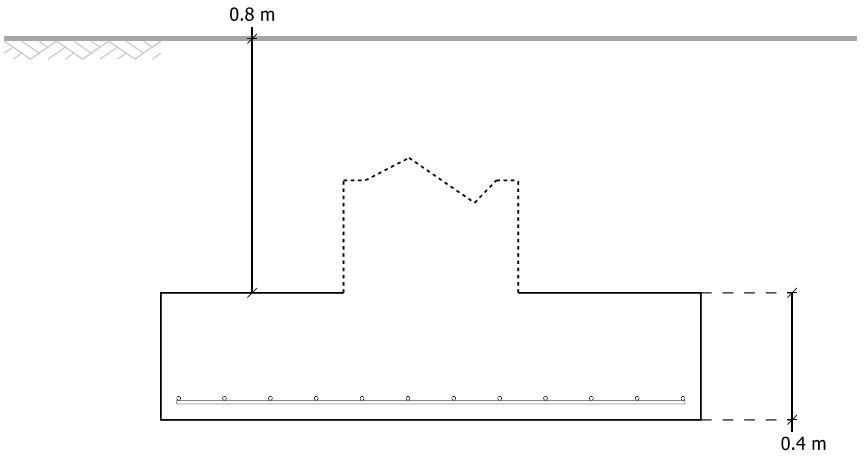
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.164

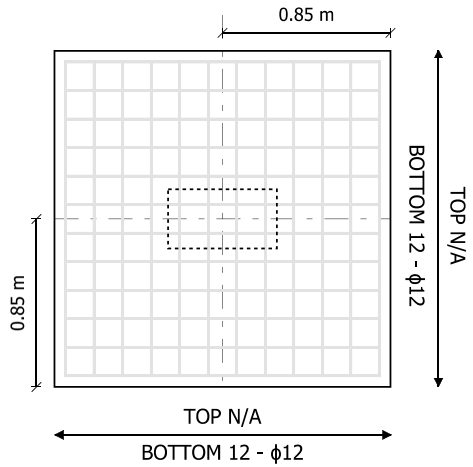
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
Foundation	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
	1.7 m	1.7 m	0.4 m	N/A	N/A	12 - ϕ12	12 - ϕ12
Pedestal	l	b	h	Main Reinforcement		Trans Reinforcement	
	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

COMBINED FOUNDATION DESIGN (ACI 318-05)

Result Summary

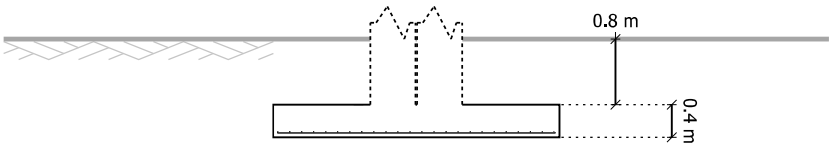
Footing No.	Left Overhang (m)	Right Overhang (m)	Length (m)	Width (m)	Thickness (m)
1	1.465	1.465	3.500	2.200	0.400

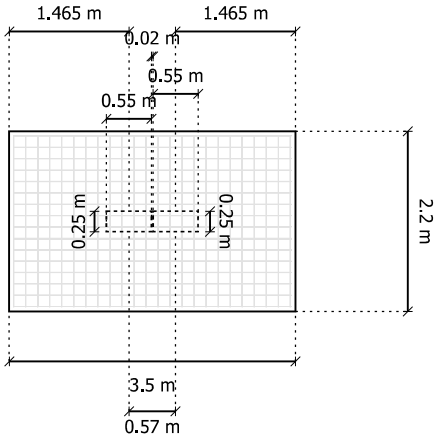
Footing No.	Footing Reinforcement			
-	Main Steel Top	Main Steel Bottom	Secondary Steel Top	Secondary Steel Bottom
1	-	16 - ϕ12	-	24 - ϕ12

FOUNDATION-CF1

Combined Footing 1

Input Data





Geometry of Footing

For Column 145

Column Dimensions

Column Shape : Rectangular

Column Length - X (D_{col}) : 0.550m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No

Pedestal Shape : N/A

Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Eccentricity

Column Offset in Transverse Direction :	0.000m

For Column 153

Column Dimensions

Column Shape :	Rectangular
Column Length - X (D_{col}) :	0.550m
Column Width - Z (B_{col}) :	0.250m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Eccentricity

Column Offset in Transverse Direction :	0.000m

Length of left overhang :	1.465m
Length of right overhang :	1.465m
Is the length of left overhang fixed?	No
Is the length of right overhang fixed?	No
Maximum length of Footing (Lo) :	400.000mm
Width of Footing (Wo) :	2.200m
Thickness of Footing (Do) :	400.000mm

Cover and Soil Properties

Pedestal Clear Cover :	50.000mm
Footing Clear Cover :	50.000mm
Unit Weight of soil :	18.000kN/m3
Soil Bearing Capacity :	350.000kN/m2
Soil Bearing Capacity Type:	Net Bearing Capacity
Soil Surcharge :	0.000kN/m2
Depth of Soil above Footing :	0.800m
Type of Depth :	Fixed Top
Depth of Water Table :	10.000m

Concrete and Rebar Properties

Unit Weight of Concrete :	24.000kN/m3
Compressive Strength of Concrete :	35.000N/mm2
Yield Strength of Steel :	415.000N/mm2
Minimum Bar Size :	#12
Maximum Bar Size :	#12
Minimum Pedestal Bar Size :	#12
Maximum Pedestal Bar Size :	#25
Minimum Bar Spacing :	100.000mm

Maximum Bar Spacing : 150.000mm

Design Calculations

Footing Size Calculations

Gross Soil Bearing Capacity = 371.60kN/m^2
Reduction of force due to buoyancy = 0.000kN

Area from initial length and width, A_o = L_o X W_o = 7.700m^2
Min. area required from bearing pressure, A_{min} = P / q_{max} = 1.369m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final footing dimensions are:

Length of footing, L : 3.500m
Width of footing, W : 2.200m
Depth of footing, Do : 0.400m
Area, A : 7.700m^2
Length of left overhang, $L_{left_overhang}$: 1.465m
Length of right overhang, $L_{right_overhang}$: 1.465m
Footing self weight : 73.917kN
Soi weight on top of footing : 106.916kN

Load Combinations

Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00
Load Combination Number	**Load Combination Title**	**Load Combination Factor**	**Soil Bearing Factor**	**Self Weight Factor**
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00

Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

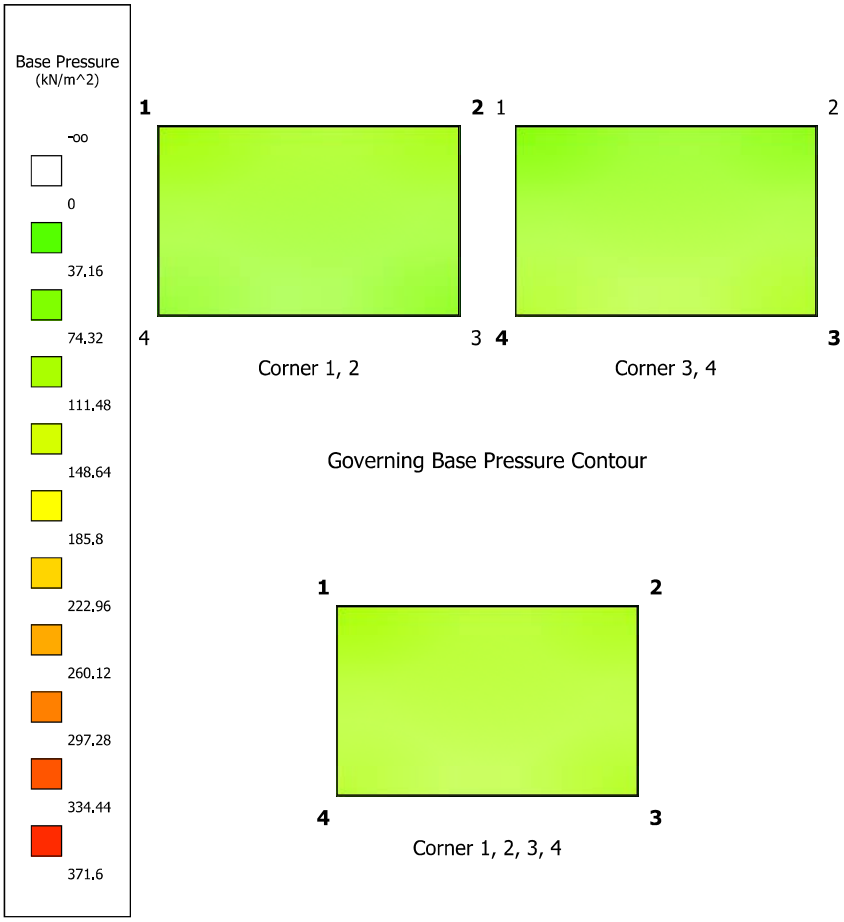
Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
-					
Column Number : 145					
10	154.780	16.812	3.325	1.436	-26.208
11	154.780	16.812	3.325	1.436	-26.208
12	161.836	18.013	3.324	1.434	-28.091
13	160.072	17.713	3.324	1.435	-27.620
14	154.047	16.848	5.486	3.378	-26.343
15	157.510	21.296	3.338	1.449	-40.251
16	159.536	18.610	3.458	1.522	-31.671
17	155.383	16.445	16.294	12.908	-23.474
18	92.153	11.283	2.173	0.978	-21.126
19	86.616	8.397	19.287	16.159	-10.197
20	95.598	14.571	2.008	0.874	-29.768
21	92.135	10.123	4.156	2.804	-15.860
22	154.065	18.008	3.503	1.553	-31.609
23	148.528	15.121	20.617	16.734	-20.681
39	158.105	16.381	3.447	1.511	-24.418
40	163.414	16.506	-9.685	-10.068	-23.482
41	90.245	8.312	2.158	0.964	-11.455
42	97.323	8.478	-15.351	-14.476	-10.208
45	152.157	15.037	3.488	1.539	-21.938
46	159.235	15.203	-14.021	-13.901	-20.691
-					
Column Number : 153					
10	155.594	-18.071	3.327	1.439	30.255
11	155.594	-18.071	3.327	1.439	30.255
12	162.689	-19.332	3.326	1.436	32.327
13	160.916	-19.017	3.327	1.437	31.809
14	154.851	-18.061	5.449	3.353	30.240
15	154.952	-16.846	3.326	1.437	26.798
16	159.138	-17.982	3.451	1.515	29.560
17	156.611	-18.343	16.281	12.896	29.586
18	90.987	-9.463	2.162	0.968	15.154
19	87.617	-9.944	19.269	16.142	15.189
20	92.714	-9.618	1.995	0.862	14.696
21	92.614	-10.832	4.119	2.778	18.139
22	153.224	-16.691	3.493	1.543	27.255
23	149.854	-17.173	20.600	16.718	27.291
39	159.928	-19.202	3.456	1.520	33.550
40	164.625	-18.398	-9.661	-10.046	29.582
41	92.040	-11.090	2.169	0.975	20.475
42	98.303	-10.018	-15.321	-14.447	15.183
45	154.278	-18.318	3.500	1.550	32.576

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
46	160.540	-17.247	-13.990	-13.872	27.285

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
-					
Column Number : 145					
24	216.692	23.537	4.655	2.011	-36.691
25	220.221	24.137	4.654	2.010	-37.632
26	220.221	24.137	4.654	2.010	-37.632
27	196.454	23.053	4.131	1.813	-38.783
28	179.260	18.070	31.657	26.199	-23.547
29	188.120	22.688	4.274	1.909	-41.034
30	179.260	18.070	31.657	26.199	-23.547
31	189.636	26.580	4.009	1.741	-51.512
32	184.688	20.226	7.077	4.498	-31.642
33	138.157	17.044	3.277	1.479	-32.230
34	129.298	12.426	30.660	25.769	-14.743
35	138.254	15.182	6.080	4.067	-23.780
36	143.202	21.536	3.011	1.310	-43.649
47	194.927	20.676	4.119	1.802	-31.046
48	196.392	18.200	-23.764	-22.817	-23.564
49	185.067	17.934	4.250	1.886	-25.559
50	196.392	18.200	-23.764	-22.817	-23.564
53	135.104	12.290	3.253	1.457	-16.755
54	146.430	12.556	-24.761	-23.247	-14.760
-					
Column Number : 153					
24	217.832	-25.300	4.658	2.014	42.356
25	221.380	-25.930	4.658	2.013	43.393
26	221.380	-25.930	4.658	2.013	43.393
27	196.169	-22.598	4.124	1.806	37.222
28	181.077	-20.878	31.628	26.172	32.600
29	186.469	-20.108	4.257	1.893	32.543
30	181.077	-20.878	31.628	26.172	32.600
31	185.796	-19.936	3.991	1.725	31.368
32	185.652	-21.671	7.024	4.461	36.285
33	136.243	-14.056	3.259	1.462	22.430
34	130.851	-14.827	30.630	25.742	22.487
35	138.973	-16.249	6.026	4.030	27.209
36	139.117	-14.514	2.993	1.293	22.292
47	197.012	-23.900	4.129	1.812	41.479
48	198.174	-20.997	-23.715	-22.771	32.590
49	188.154	-22.711	4.268	1.903	41.057
50	198.174	-20.997	-23.715	-22.771	32.590
53	137.929	-16.659	3.270	1.473	30.944
54	147.948	-14.945	-24.713	-23.202	22.478

Pressures at 4 Corners

Governing Base Pressure Contour for Each Corner



Load Case	Pressure at corner 1 (q_1) (kN/m^2)	Pressure at corner 2 (q_2) (kN/m^2)	Pressure at corner 3 (q_3) (kN/m^2)	Pressure at corner 4 (q_4) (kN/m^2)	Area of footing in uplift (A_u) (sq. m)
46	**80.3869**	77.2524	49.6415	52.7759	0.000
46	80.3869	**77.2524**	49.6415	52.7759	0.000
23	46.1180	42.9775	**78.3536**	81.4941	0.000
23	46.1180	42.9775	78.3536	**81.4941**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m^2)	Pressure at corner 2 (q_2) (kN/m^2)	Pressure at corner 3 (q_3) (kN/m^2)	Pressure at corner 4 (q_4) (kN/m^2)
46	**80.3869**	77.2524	49.6415	52.7759
46	80.3869	**77.2524**	49.6415	52.7759
23	46.1180	42.9775	**78.3536**	81.4941
23	46.1180	42.9775	78.3536	**81.4941**

Stability Check

Load Case	Shear X (kN)	Shear Z (kN)	Resultant Shear (kN)	Resisting Sliding Force (kN)	Ratio X	Ratio Z	Resultant Ratio	Required FOS
10	-1.259	6.652	6.770	196.483	156.021	29.537	29.022	1.500
11	-1.259	6.652	6.770	196.483	156.021	29.537	29.022	1.500
12	-1.318	6.651	6.780	202.144	153.315	30.395	29.815	1.500
13	-1.304	6.651	6.778	200.729	153.969	30.180	29.617	1.500
14	-1.213	10.936	11.003	195.893	161.507	17.913	17.804	1.500
15	4.449	6.664	8.013	197.318	44.349	29.610	24.625	1.500
16	0.628	6.909	6.937	199.803	318.159	28.921	28.802	1.500
17	-1.898	32.575	32.630	197.131	103.870	6.052	6.041	1.500
18	1.820	4.335	4.701	145.589	79.994	33.587	30.968	1.500
19	-1.548	38.556	38.587	142.026	91.759	3.684	3.681	1.500
20	4.953	4.003	6.368	147.658	29.812	36.885	23.186	1.500
21	-0.709	8.275	8.305	146.233	206.201	17.672	17.608	1.500
22	1.316	6.996	7.118	195.249	148.336	27.910	27.429	1.500
23	-2.052	41.217	41.268	191.686	93.435	4.651	4.645	1.500
39	-2.820	6.903	7.457	199.547	70.752	28.909	26.761	1.500
40	-1.892	-19.346	19.439	203.549	107.566	10.521	10.471	1.500
41	-2.778	4.327	5.142	145.247	52.288	33.570	28.249	1.500
42	-1.540	-30.672	30.710	150.584	97.754	4.910	4.903	1.500
45	-3.282	6.988	7.720	194.907	59.395	27.894	25.248	1.500
46	-2.044	-28.011	28.085	200.244	97.958	7.149	7.130	1.500

Check for stability against overturning (Moments printed against Local axis)

Load Case	Moment X (kNm)	Moment Z (kNm)	Resisting Moment X (kNm)	Resisting Moment Z (kNm)	Ratio X	Ratio Z	Required FOS
10	5.536	4.319	540.319	859.599	97.604	199.044	1.500
11	5.536	4.319	540.319	859.599	97.604	199.044	1.500
12	5.530	4.521	555.885	884.363	100.515	195.615	1.500
13	5.532	4.470	551.994	878.172	99.787	196.443	1.500
14	11.105	4.153	538.695	857.015	48.508	206.338	1.500
15	5.552	-14.504	542.616	863.252	97.737	59.519	1.500
16	5.801	-2.249	549.448	874.122	94.722	388.603	1.500
17	38.834	6.521	542.100	862.432	13.960	132.252	1.500
18	3.680	-6.368	400.363	636.941	108.793	100.014	1.500
19	47.725	5.326	390.565	621.354	8.184	116.674	1.500
20	3.338	-16.231	406.053	645.994	121.665	39.800	1.500
21	8.891	2.426	402.133	639.756	45.229	263.708	1.500
22	5.894	-4.641	536.925	854.199	91.092	184.054	1.500
23	49.939	7.053	527.128	838.612	10.556	118.901	1.500
39	5.793	9.741	548.743	873.000	94.731	89.618	1.500
40	-27.853	6.511	559.749	890.510	20.097	136.772	1.500
41	3.669	9.619	399.423	635.445	108.854	66.061	1.500
42	-41.192	5.312	414.098	658.792	10.053	124.021	1.500
45	5.884	11.347	535.985	852.704	91.097	75.151	1.500
46	-38.977	7.039	550.660	876.050	14.128	124.450	1.500

Calculations of Footing Thickness

Footing thickness is calculated based on the ultimate load cases

Punching Shear Check

For Column 145

Critical Load case for Punching Shear Check : 25

Total Footing Depth, D_o = 0.400m

Calculated Effective Depth, d_{eff} = 0.344m

For rectangular column, β_c = B_{col} / D_{col} : 2.200

Considering the particular column as interior column, Slab Edge Factor α_s : 40.0

Effective depth, d_{eff}, increased until $0.75*V_c \geq$ Punching Shear Force

Punchng Shear Force, Vu = 189.920kN

From ACI Cl.11.12.2.1, b_c for column 2.976m

Equation 11-33, V_{c1} $\left(2+\frac{4}{\beta_c}\right)\times b_o \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 1920.178kN

Equation 11-34, V_{c2} = $\left(\frac{\alpha_s \times d}{b_o}+2\right)\times \lambda \times \sqrt{f_c} \times b_o \times d$ = 3331.062kN

Equation 11-35, V_{c3} = $4 \times b_o \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 2011.615kN

Punching shear strength, V_c = 0.75 x minimum of (V_{c1},V_{c2},V_{c3}) = 1440.134kN

$0.75 * V_c > V_u$ hence, OK

For Column 153

Critical Load case for Punching Shear Check : 25

Total Footing Depth, D_o = 0.400m

Calculated Effective Depth, d_{eff} = 0.344m

For rectangular column, β_c = B_{col} / D_{col} : 2.200

Considering the particular column as interior column, Slab Edge Factor α_s : 40.0

Effective depth, d_{eff}, increased until $0.75*V_c \geq$ Punching Shear Force

Punchng Shear Force, Vu = 191.316kN

From ACI Cl.11.12.2.1, b_c for column 2.976m

Equation 11-33, V_{c1} $\left(2+\frac{4}{\beta_c}\right)\times b_o \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 1920.178kN

Equation 11-34, V_{c2} = $\left(\frac{\alpha_s \times d}{b_o}+2\right)\times \lambda \times \sqrt{f_c} \times b_o \times d$ = 3331.062kN

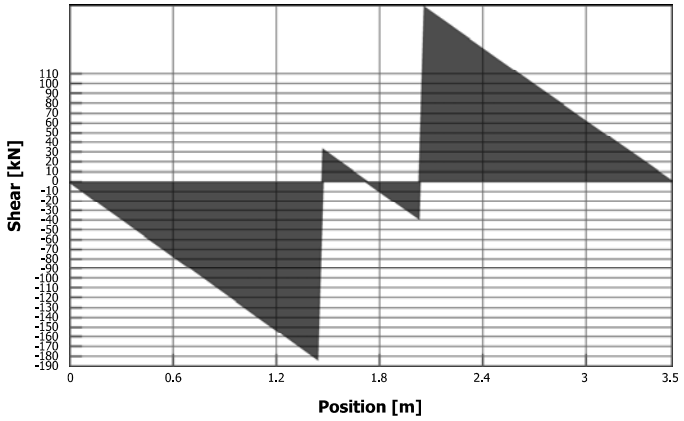
Equation 11-35, V_{c3} = $4 \times b_o \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 2011.615kN

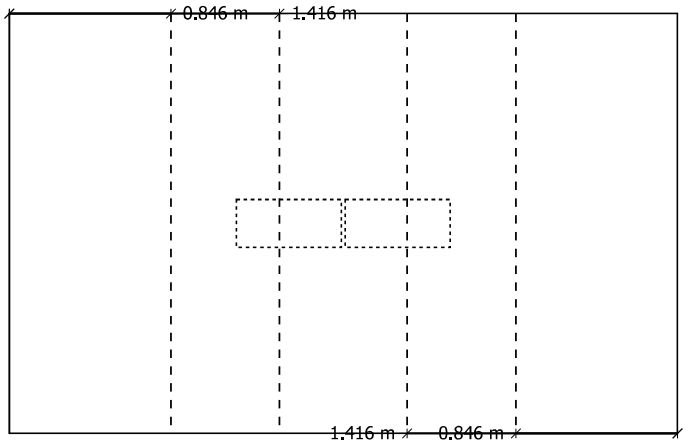
Punching shear strength, V_c = 0.75 x minimum of (V_{c1},V_{c2},V_{c3}) = 1440.134kN

$0.75 * V_c > V_u$ hence, OK

One-Way Shear

Shear Diagram





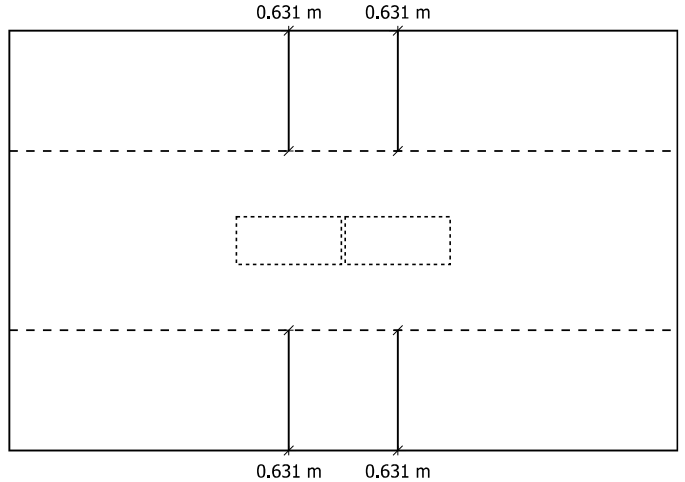
Shear Plane Parallel to Foundation Width

Critical load case for maximum shear force along the length of footing : 25

Critical Shear force, V_u	For the critical load case:	179.595kN
Point of occurance of V_u	Critical one-way shear position:	1.416m
From ACI Cl.11.3.1.1, V_c =	$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ =	743.541kN

0.75 x V_c = 557.655kN

Since 0.75 * V_c > V_u hence, OK



Shear Plane Parallel to Foundation Length

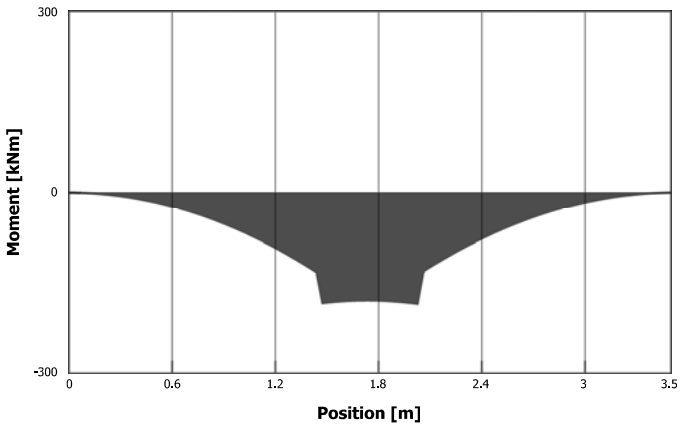
Critical load case for maximum shear force along the width of footing : 48

Critical Shear force,V_u	For the critical load case:	152.021kN
Point of occurance of V_u	Critical one-way shear position:	0.631m
From ACI Cl.11.3.1.1, V_c =	$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ =	1182.906kN
	0.75 x V_c =	887.179kN
	Since 0.75 * V_c > V_u	hence, OK

Design for Flexure

Design of flexure

Moment Diagram



Bottom Reinforcement

Critical load case :		25
Required Effective Depth :		0.344m
β_1, from ACI Cl.10.2.7.3	max[0.85-0.05x(Fc-4),0.65]=	0.7962
From ACI Cl. 10.3.2, ρ_{bal}	$0.85 \times \beta_1 \times F_c' \times \frac{87}{[f_y \times (87 + F_y)]}$ =	0.03374
From ACI Cl. 10.3.3, ρ_{max}	$0.75 \times \rho_{bal}$ =	0.02530
From ACI Cl. 7.12.2, ρ_{min}	$\max\left(0.0018 \cdot \frac{60ksi}{F_y}, 0.0014\right)$ =	0.00200
Modular Ratio,m	$\frac{F_y}{(0.85 \times F_c')}$ =	13.9496

Ultimate Moment (Mz) :	180.823kNm
Point of occurrence of the ultimate moment along the length of footing :	1.760m
Nominal Moment Capacity :	200.914kNm
Required ρ (based on effective depth) :	0.0023
ρ x d_{eff} / Depth (based on gross depth) :	0.0020
Area of main steel required, A_s = ρ * W * deff :	1760.000mm2

Distribution Reinforcement

Critical load case :	48		
Critical Moment for distribution steel (Mx) :		113.1205	kNm
Nominal moment Capacity :	125.6894		kNm
Point of occurance of the critical moment along length:	1.2250		m
Required ρ (based on effective depth):	0.0024		
ρ x d_{eff} / Depth (based on gross depth) :	0.0020		
Area of distribution steel required, A_s = ρ * L * deff :	1760.000		mm2

Provided Reinforcement

Based on spacing reinforcement increment; provided reinforcement is

Main bar no. for bottom Reinforcement:	#12	
Spacing of bottom reinforcement bar :	140.000	mm

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 140mm o.c.

Distribution bar no. (Bottom):	#12	
Spacing of distribution bars (Bottom):	147.826	mm

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 145mm o.c.

Based on spacing reinforcement increment; provided reinforcement is

Pedestal Design

Print Calculation Sheet

COMBINED FOUNDATION DESIGN (ACI 318-05)

Result Summary

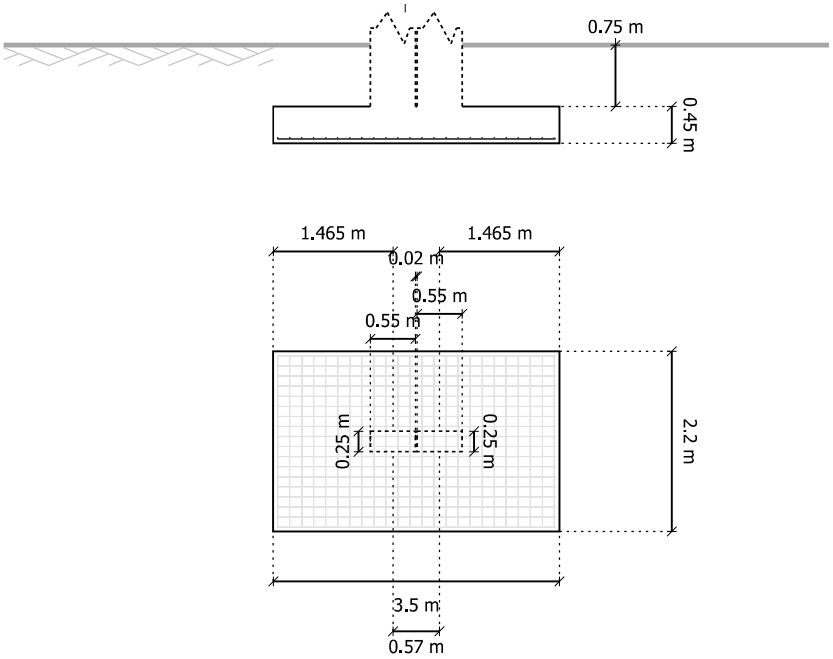
Footing No.	Left Overhang (m)	Right Overhang (m)	Length (m)	Width (m)	Thickness (m)
1	1.465	1.465	3.500	2.200	0.450

Footing No.	Footing Reinforcement			
-	Main Steel Top	Main Steel Bottom	Secondary Steel Top	Secondary Steel Bottom
1	-	18 - ϕ12	-	24 - ϕ12

FOUNDATION-CF2

Combined Footing 1

Input Data



Geometry of Footing

For Column 145

Column Dimensions

Column Shape : Rectangular

Column Length - X (D_{col}) : 0.550m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No

Pedestal Shape : N/A

Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Eccentricity

Column Offset in Transverse Direction : 0.000m

For Column 153

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.550m
Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Eccentricity

Column Offset in Transverse Direction : 0.000m

Length of left overhang : 1.465m
Length of right overhang : 1.465m
Is the length of left overhang fixed? No
Is the length of right overhang fixed? No
Maximum length of Footing (Lo) : 450.000mm
Width of Footing (Wo) : 2.200m
Thickness of Footing (Do) : 450.000mm

Cover and Soil Properties

Pedestal Clear Cover : 50.000mm
Footing Clear Cover : 50.000mm
Unit Weight of soil : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 0.750m
Type of Depth : Fixed Top
Depth of Water Table : 10.000m

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Compressive Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 415.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Minimum Pedestal Bar Size : #12
Maximum Pedestal Bar Size : #25
Minimum Bar Spacing : 100.000mm

Maximum Bar Spacing : 150.000mm

Design Calculations

Footing Size Calculations

Gross Soil Bearing Capacity = 371.60kN/m^2
Reduction of force due to buoyancy = 0.000kN

Area from initial length and width, A_o = L_o X W_o = 7.700m^2
Min. area required from bearing pressure, A_{min} = P / q_{max} = 1.376m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final footing dimensions are:

Length of footing, L : 3.500m
Width of footing, W : 2.200m
Depth of footing, Do : 0.450m
Area, A : 7.700m^2
Length of left overhang, $L_{left_overhang}$: 1.465m
Length of right overhang, $L_{right_overhang}$: 1.465m
Footing self weight : 83.157kN
Soi weight on top of footing : 100.234kN

Load Combinations

Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00
Load Combination Number	**Load Combination Title**	**Load Combination Factor**	**Soil Bearing Factor**	**Self Weight Factor**
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00

Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

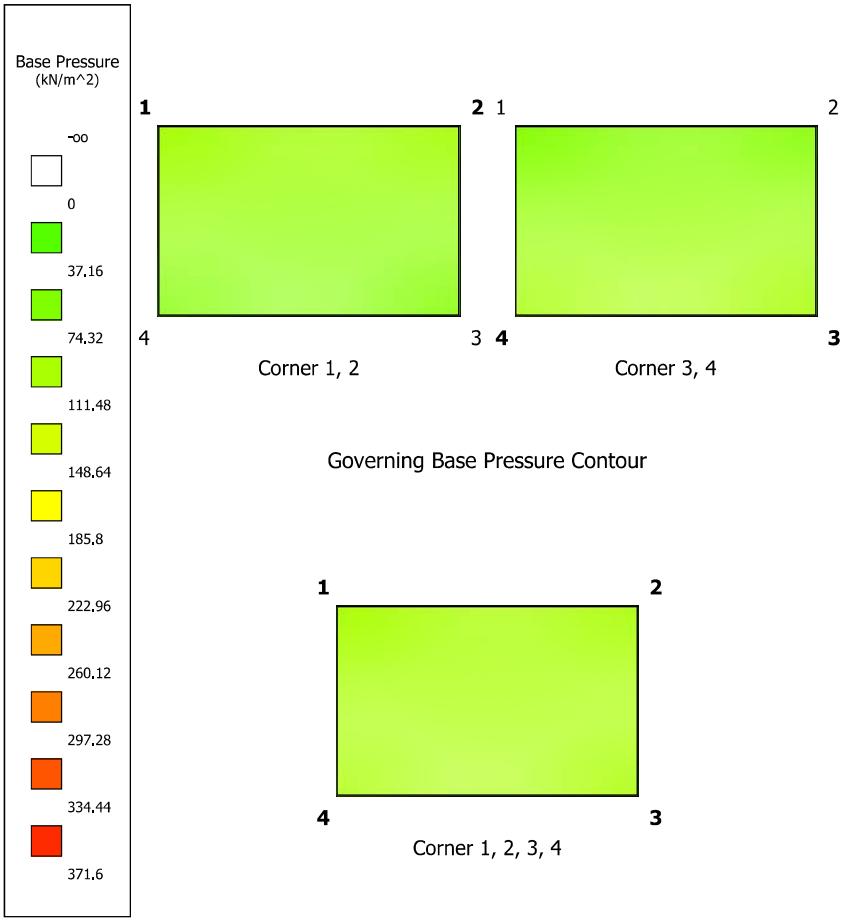
Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
-					
Column Number : 145					
10	154.780	16.812	3.325	1.436	-26.208
11	154.780	16.812	3.325	1.436	-26.208
12	161.836	18.013	3.324	1.434	-28.091
13	160.072	17.713	3.324	1.435	-27.620
14	154.047	16.848	5.486	3.378	-26.343
15	157.510	21.296	3.338	1.449	-40.251
16	159.536	18.610	3.458	1.522	-31.671
17	155.383	16.445	16.294	12.908	-23.474
18	92.153	11.283	2.173	0.978	-21.126
19	86.616	8.397	19.287	16.159	-10.197
20	95.598	14.571	2.008	0.874	-29.768
21	92.135	10.123	4.156	2.804	-15.860
22	154.065	18.008	3.503	1.553	-31.609
23	148.528	15.121	20.617	16.734	-20.681
39	158.105	16.381	3.447	1.511	-24.418
40	163.414	16.506	-9.685	-10.068	-23.482
41	90.245	8.312	2.158	0.964	-11.455
42	97.323	8.478	-15.351	-14.476	-10.208
45	152.157	15.037	3.488	1.539	-21.938
46	159.235	15.203	-14.021	-13.901	-20.691
-					
Column Number : 153					
10	155.594	-18.071	3.327	1.439	30.255
11	155.594	-18.071	3.327	1.439	30.255
12	162.689	-19.332	3.326	1.436	32.327
13	160.916	-19.017	3.327	1.437	31.809
14	154.851	-18.061	5.449	3.353	30.240
15	154.952	-16.846	3.326	1.437	26.798
16	159.138	-17.982	3.451	1.515	29.560
17	156.611	-18.343	16.281	12.896	29.586
18	90.987	-9.463	2.162	0.968	15.154
19	87.617	-9.944	19.269	16.142	15.189
20	92.714	-9.618	1.995	0.862	14.696
21	92.614	-10.832	4.119	2.778	18.139
22	153.224	-16.691	3.493	1.543	27.255
23	149.854	-17.173	20.600	16.718	27.291
39	159.928	-19.202	3.456	1.520	33.550
40	164.625	-18.398	-9.661	-10.046	29.582
41	92.040	-11.090	2.169	0.975	20.475
42	98.303	-10.018	-15.321	-14.447	15.183
45	154.278	-18.318	3.500	1.550	32.576

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
46	160.540	-17.247	-13.990	-13.872	27.285

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
-					
Column Number : 145					
24	216.692	23.537	4.655	2.011	-36.691
25	220.221	24.137	4.654	2.010	-37.632
26	220.221	24.137	4.654	2.010	-37.632
27	196.454	23.053	4.131	1.813	-38.783
28	179.260	18.070	31.657	26.199	-23.547
29	188.120	22.688	4.274	1.909	-41.034
30	179.260	18.070	31.657	26.199	-23.547
31	189.636	26.580	4.009	1.741	-51.512
32	184.688	20.226	7.077	4.498	-31.642
33	138.157	17.044	3.277	1.479	-32.230
34	129.298	12.426	30.660	25.769	-14.743
35	138.254	15.182	6.080	4.067	-23.780
36	143.202	21.536	3.011	1.310	-43.649
47	194.927	20.676	4.119	1.802	-31.046
48	196.392	18.200	-23.764	-22.817	-23.564
49	185.067	17.934	4.250	1.886	-25.559
50	196.392	18.200	-23.764	-22.817	-23.564
53	135.104	12.290	3.253	1.457	-16.755
54	146.430	12.556	-24.761	-23.247	-14.760
-					
Column Number : 153					
24	217.832	-25.300	4.658	2.014	42.356
25	221.380	-25.930	4.658	2.013	43.393
26	221.380	-25.930	4.658	2.013	43.393
27	196.169	-22.598	4.124	1.806	37.222
28	181.077	-20.878	31.628	26.172	32.600
29	186.469	-20.108	4.257	1.893	32.543
30	181.077	-20.878	31.628	26.172	32.600
31	185.796	-19.936	3.991	1.725	31.368
32	185.652	-21.671	7.024	4.461	36.285
33	136.243	-14.056	3.259	1.462	22.430
34	130.851	-14.827	30.630	25.742	22.487
35	138.973	-16.249	6.026	4.030	27.209
36	139.117	-14.514	2.993	1.293	22.292
47	197.012	-23.900	4.129	1.812	41.479
48	198.174	-20.997	-23.715	-22.771	32.590
49	188.154	-22.711	4.268	1.903	41.057
50	198.174	-20.997	-23.715	-22.771	32.590
53	137.929	-16.659	3.270	1.473	30.944
54	147.948	-14.945	-24.713	-23.202	22.478

Pressures at 4 Corners

Governing Base Pressure Contour for Each Corner



Load Case	Pressure at corner 1 (q_1) (kN/m^2)	Pressure at corner 2 (q_2) (kN/m^2)	Pressure at corner 3 (q_3) (kN/m^2)	Pressure at corner 4 (q_4) (kN/m^2)	Area of footing in uplift (A_u) (sq. m)
46	**81.2378**	78.0578	49.4548	52.6348	0.000
46	81.2378	**78.0578**	49.4548	52.6348	0.000
23	45.7431	42.5568	**79.3928**	82.5790	0.000
23	45.7431	42.5568	79.3928	**82.5790**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m^2)	Pressure at corner 2 (q_2) (kN/m^2)	Pressure at corner 3 (q_3) (kN/m^2)	Pressure at corner 4 (q_4) (kN/m^2)
46	**81.2378**	78.0578	49.4548	52.6348
46	81.2378	**78.0578**	49.4548	52.6348
23	45.7431	42.5568	**79.3928**	82.5790
23	45.7431	42.5568	79.3928	**82.5790**

Stability Check

Load Case	Shear X (kN)	Shear Z (kN)	Resultant Shear (kN)	Resisting Sliding Force (kN)	Ratio X	Ratio Z	Resultant Ratio	Required FOS
10	-1.259	6.652	6.770	197.506	156.834	29.691	29.173	1.500
11	-1.259	6.652	6.770	197.506	156.834	29.691	29.173	1.500
12	-1.318	6.651	6.780	203.167	154.091	30.549	29.965	1.500
13	-1.304	6.651	6.778	201.752	154.753	30.334	29.768	1.500
14	-1.213	10.936	11.003	196.916	162.350	18.007	17.897	1.500
15	4.449	6.664	8.013	198.341	44.579	29.763	24.753	1.500
16	0.628	6.909	6.937	200.826	319.787	29.069	28.950	1.500
17	-1.898	32.575	32.630	198.154	104.409	6.083	6.073	1.500
18	1.820	4.335	4.701	146.612	80.556	33.823	31.186	1.500
19	-1.548	38.556	38.587	143.049	92.420	3.710	3.707	1.500
20	4.953	4.003	6.368	148.681	30.019	37.141	23.346	1.500
21	-0.709	8.275	8.305	147.256	207.644	17.796	17.731	1.500
22	1.316	6.996	7.118	196.272	149.113	28.057	27.573	1.500
23	-2.052	41.217	41.268	192.709	93.934	4.675	4.670	1.500
39	-2.820	6.903	7.457	200.570	71.115	29.057	26.899	1.500
40	-1.892	-19.346	19.439	204.572	108.106	10.574	10.524	1.500
41	-2.778	4.327	5.142	146.270	52.656	33.807	28.448	1.500
42	-1.540	-30.672	30.710	151.607	98.418	4.943	4.937	1.500
45	-3.282	6.988	7.720	195.930	59.706	28.040	25.380	1.500
46	-2.044	-28.011	28.085	201.267	98.459	7.185	7.166	1.500

Check for stability against overturning (Moments printed against Local axis)

Load Case	Moment X (kNm)	Moment Z (kNm)	Resisting Moment X (kNm)	Resisting Moment Z (kNm)	Ratio X	Ratio Z	Required FOS
10	5.869	4.382	543.132	864.074	92.552	197.205	1.500
11	5.869	4.382	543.132	864.074	92.552	197.205	1.500
12	5.863	4.587	558.698	888.838	95.294	193.780	1.500
13	5.864	4.536	554.807	882.647	94.608	194.607	1.500
14	11.652	4.214	541.508	861.490	46.473	204.430	1.500
15	5.885	-14.726	545.429	867.727	92.682	58.924	1.500
16	6.146	-2.281	552.261	878.597	89.856	385.216	1.500
17	40.463	6.616	544.913	866.907	13.467	131.031	1.500
18	3.897	-6.459	403.176	641.416	103.464	99.298	1.500
19	49.653	5.403	393.378	625.829	7.923	115.831	1.500
20	3.538	-16.479	408.866	650.469	115.577	39.473	1.500
21	9.305	2.461	404.946	644.232	43.521	261.728	1.500
22	6.244	-4.707	539.738	858.675	86.439	182.431	1.500
23	52.000	7.156	529.941	843.088	10.191	117.822	1.500
39	6.138	9.882	551.556	877.476	89.863	88.793	1.500
40	-28.820	6.606	562.562	894.985	19.520	135.490	1.500
41	3.886	9.758	402.236	639.921	103.518	65.579	1.500
42	-42.725	5.389	416.911	663.267	9.758	123.079	1.500
45	6.233	11.511	538.798	857.179	86.442	74.468	1.500
46	-40.378	7.142	553.473	880.525	13.708	123.296	1.500

Calculations of Footing Thickness

Footing thickness is calculated based on the ultimate load cases

Punching Shear Check

For Column 145

Critical Load case for Punching Shear Check : 25

Total Footing Depth, D_o = 0.450m

Calculated Effective Depth, d_{eff} = 0.394m

For rectangular column, β_c = B_{col} / D_{col} : 2.200

Considering the particular column as interior column, Slab Edge Factor α_s : 40.0

Effective depth, d_{eff}, increased until $0.75*V_c \geq$ Punching Shear Force

Punchng Shear Force, Vu = 185.511kN

From ACI Cl.11.12.2.1, b_c for column 3.176m

Equation 11-33, V_{c1} $\left(2+\frac{4}{\beta_c}\right)\times b_o \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 2347.075kN

Equation 11-34, V_{c2} = $\left(\frac{\alpha_s \times d}{b_o}+2\right)\times \lambda \times \sqrt{f_c} \times b_o \times d$ = 4279.745kN

Equation 11-35, V_{c3} = $4 \times b_o \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 2458.840kN

Punching shear strength, V_c = 0.75 x minimum of (V_{c1},V_{c2},V_{c3}) = 1760.306kN

$0.75 * V_c > V_u$ hence, OK

For Column 153

Critical Load case for Punching Shear Check : 25

Total Footing Depth, D_o = 0.450m

Calculated Effective Depth, d_{eff} = 0.394m

For rectangular column, β_c = B_{col} / D_{col} : 2.200

Considering the particular column as interior column, Slab Edge Factor α_s : 40.0

Effective depth, d_{eff}, increased until $0.75*V_c \geq$ Punching Shear Force

Punchng Shear Force, Vu = 186.945kN

From ACI Cl.11.12.2.1, b_c for column 3.176m

Equation 11-33, V_{c1} $\left(2+\frac{4}{\beta_c}\right)\times b_o \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 2347.075kN

Equation 11-34, V_{c2} = $\left(\frac{\alpha_s \times d}{b_o}+2\right)\times \lambda \times \sqrt{f_c} \times b_o \times d$ = 4279.745kN

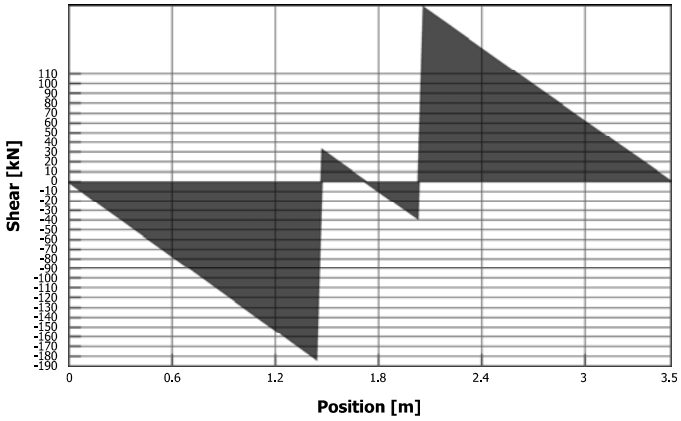
Equation 11-35, V_{c3} = $4 \times b_o \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 2458.840kN

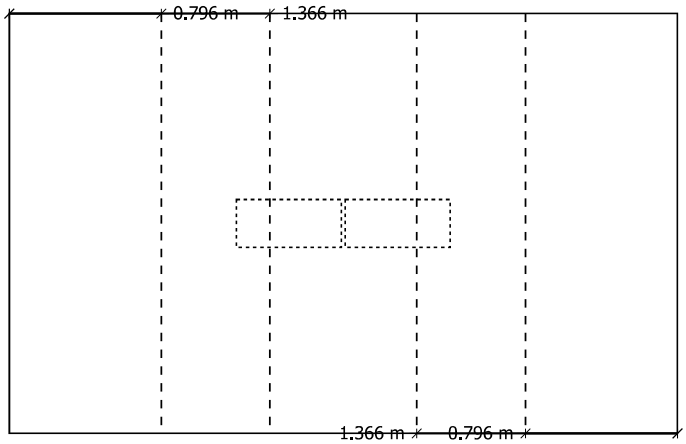
Punching shear strength, V_c = 0.75 x minimum of (V_{c1},V_{c2},V_{c3}) = 1760.306kN

$0.75 * V_c > V_u$ hence, OK

One-Way Shear

Shear Diagram

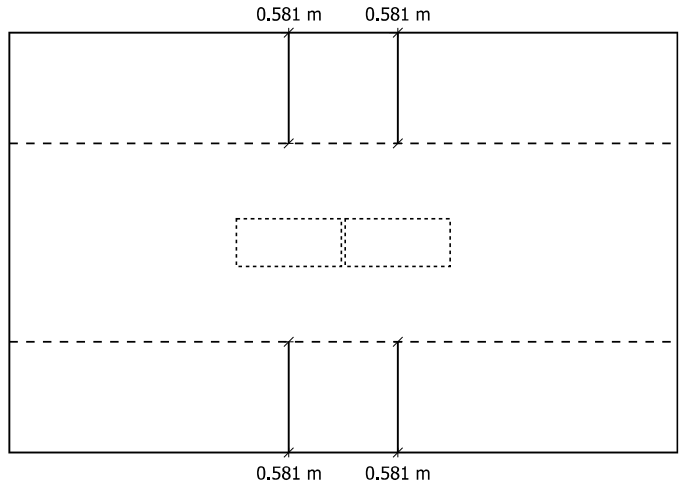




Shear Plane Parallel to Foundation Width

Critical load case for maximum shear force along the length of footing : 25

Critical Shear force, V_u	For the critical load case:	173.446kN
Point of occurance of V_u	Critical one-way shear position:	1.366m
From ACI Cl.11.3.1.1, V_c =	$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ =	851.613kN
	0.75 x V_c =	638.710kN
	Since 0.75 * V_c > V_u	hence, OK



Shear Plane Parallel to Foundation Length

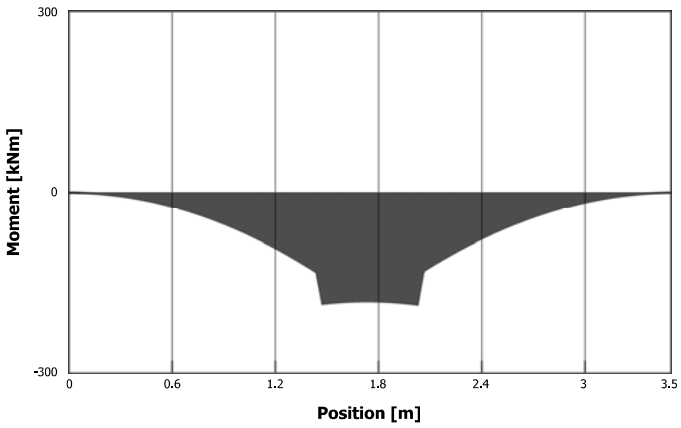
Critical load case for maximum shear force along the width of footing : 48

Critical Shear force,V_u	For the critical load case:	142.422kN
Point of occurance of V_u	Critical one-way shear position:	0.581m
From ACI Cl.11.3.1.1, V_c =	$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ =	1354.840kN
	0.75 x V_c =	1016.130kN
	Since 0.75 * V_c > V_u	hence, OK

Design for Flexure

Design of flexure

Moment Diagram



Bottom Reinforcement

Critical load case :		25
Required Effective Depth :		0.394m
β_1, from ACI Cl.10.2.7.3	max[0.85-0.05x(Fc-4),0.65]=	0.7962
From ACI Cl. 10.3.2, ρ_{bal}	$0.85 \times \beta_1 \times F_c' \times \frac{87}{[f_y \times (87 + F_y)]}$ =	0.03374
From ACI Cl. 10.3.3, ρ_{max}	$0.75 \times \rho_{bal}$ =	0.02530
From ACI Cl. 7.12.2, ρ_{min}	$\max\left(0.0018 \cdot \frac{60ksi}{F_y}, 0.0014\right)$ =	0.00200
Modular Ratio,m	$\frac{F_y}{(0.85 \times F_c')}$ =	13.9496

Ultimate Moment (Mz) :	182.075kNm
Point of occurrence of the ultimate moment along the length of footing :	1.760m
Nominal Moment Capacity :	202.306kNm
Required ρ (based on effective depth) :	0.0023
ρ x d_{eff} / Depth (based on gross depth) :	0.0020
Area of main steel required, A_s = ρ * W * deff :	1980.000mm2

Distribution Reinforcement

Critical load case :	48		
Critical Moment for distribution steel (Mx) :		114.0385	kNm
Nominal moment Capacity :	126.7094		kNm
Point of occurance of the critical moment along length:	1.2250		m
Required ρ (based on effective depth):	0.0024		
ρ x d_{eff} / Depth (based on gross depth) :	0.0020		
Area of distribution steel required, A_s = ρ * L * deff :	1980.000		mm2

Provided Reinforcement

Based on spacing reinforcement increment; provided reinforcement is

Main bar no. for bottom Reinforcement:	#12	
Spacing of bottom reinforcement bar :	123.529	mm

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 120mm o.c.

Distribution bar no. (Bottom):	#12	
Spacing of distribution bars (Bottom):	147.826	mm

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 145mm o.c.

Based on spacing reinforcement increment; provided reinforcement is

Pedestal Design

Print Calculation Sheet